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HAZOPCT: A HAZOP analysis completeness tool based on knowledge graph reasoning

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ABSTRACT

Hazard and Operability (HAZOP) analysis is widely recognized as a cornerstone of risk assessment in the chemical industry. However, the reliance on manual processes makes HAZOP analysis time-consuming, labor-intensive, and subject to variability, leading to inconsistent report quality and increased accident risks. To address these challenges, this paper introduces a HAZOP analysis Completeness Tool (HAZOPCT), which helps to review process designs for enhancing the completeness of HAZOP analysis. Initially, a HAZOP knowledge graph is constructed automatically to integrate multi-source process safety information based on a predefined ontology. Subsequently, HAZOPCT is employed to perform reasoning over the knowledge graph, thereby generating recommended conclusions from similar cases for completing HAZOP analysis. A case study focusing on the sulfur recovery process demonstrates the ability of HAZOPCT to efficiently identify hazards and operability issues that may have been overlooked in original HAZOP analysis tables. Additionally, ablation studies validate the critical role of semantic information and graph attention mechanisms in improving reasoning performance. The generated attention weight network further enhances the interpretability of HAZOPCT, offering insights into the automated reasoning process and contributing to a more transparent and reliable HAZOP analysis.

1. Introduction

Process safety risk assessment is essential for safeguarding employees, the environment, and communities from risks linked to chemical process incidents (Ab Rahim et al., 2024). Hazard and Operability (HAZOP) analysis, a cornerstone methodology in risk assessment practices, systematically reviews process designs to identify all potential deviations that may result in operational risks, and recommends controls to reduce risks to levels that are as low as reasonably practicable.

Traditional HAZOP analysis relies on multidisciplinary teams comprising five to eight experts who conduct collaborative reviews (American Institute of Chemical Engineers, 2008). Given that the information utilized in HAZOP analysis is extensive and that every possible deviation needs to be analyzed, the process is both time-consuming and resource-intensive. Furthermore, disparities in the experience, domain knowledge, or even interpretations of HAZOP terminology among HAZOP analysis teams may lead to inconsistencies in the quality of HAZOP analysis reports (Baybutt, 2015). Unfortunately, poor-quality HAZOP analysis reports have been identified as critical contributors to many safety incidents within the chemical industry. Table 1 provides information of safety incidents that happened caused by deficiencies in HAZOP analysis reports as a contributing factor. The accident investigation reports indicate that the HAZOP analyses conducted during the design or maintenance phases at these factories suffered from inadequate data preparation and incomplete analysis. These incidents underscore the urgent need for an advanced approach to identify and rectify deficiencies in HAZOP analysis reports.

To address the limitations arising from subjective brainstorming sessions and varying levels of expertise among the HAZOP teams, research on intelligent HAZOP analysis commenced internationally in the 1980s (Single et al., 2019). Fig. 1 illustrates the evolutionary path of representative HAZOP tools or methods to date. Parmar and Lees (1987) were among the pioneers in introducing a rule-based modeling approach to analyze fault propagation pathways arising from deviations in process parameters. They demonstrated the effectiveness of automated HAZOP analyses in oil-water separation systems. One year later, Heino et al. (1988) developed an expert system named HAZOPEX, which provides expertise of typical hazards and operability issues in a process environment and a mechanism for logical inference based on the expertise. In subsequent years, several intelligent HAZOP analysis systems were

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Table 1
A summary of the safety incidents caused in part by deficiencies in HAZOP.

Incident	Date	Outcome	Causal Factors Related to HAZOP Quality
Flixborough Explosion (Grande-Bretagne, 1975)	Jun. 1, 1974	Loss of life (28 deaths, 36 injuries) and severe material damage	• A plant modification occurred without a full assessment of the potential consequences • No drawing of the proposed modification was produced
Bhopal Gas Leak (Lees, 1996)	Dec. 3, 1984	Mass causalities (About 2000 deaths, 10000 injuries) and severe material damage	• No information about the hazardous nature of methyl isocyanate was available • Little research has been conducted about runaway reactions occurred in a storage tank
Hydrocracker Explosion and Fire at BP Oil (Great Britain, 1989)	Mar. 22, 1987	Loss of life (1 death) and severe material damage	• The low-low limit alarm and the corresponding trip system on the high-pressure separator had been disconnected before a plant modification was made
Explosion and Fires at the Texaco Refinery (Health and Executive, 1997)	Jul. 24, 1994	Severe material damage	• A plant modification was carried out without a completed assessment of the potential consequences
Explosion and Fires at the Longford Refining Plant (Hopkins, 2000)	Sep. 25, 1998	Loss of life (2 deaths, 8 injuries) and severe material damage	• HAZOP analyses had been conducted for some aspects of risk but not for all. Therefore, the risk of embrittlement of processing equipment when exposed to super-chilled conditions was overlooked
Explosion at Sinopec Jiujiang Refinery (SINOPEC Jiujiang Petrochemical Co, 2018)	Mar. 12, 2018	Loss of life (2 deaths, 1 injury) and severe material damage	• HAZOP analyses failed to sufficiently address risks of high-pressure gas leaking to the low-pressure side due to check valves failing to close at the hydrogen feed pump outlet
Fires at Sinopec Maoming Refinery (SINOPEC Maoming Petrochemical Co, 2022)	Mar. 30, 2022	Severe material damage	• HAZOP analyses failed to sufficiently address risks related to the wild fluctuation of dilution vapor flow rate, leading to the absence of interlocking shutdown systems
Explosion and Fires at Sinopec Shanghai Refinery (SINOPEC Shanghai Petrochemical Co, 2023)	Jun. 18, 2022	Loss of life (1 death, 1 injury) and severe material damage	• HAZOP analyses failed to sufficiently address risks related to low liquid levels at the bottom of a refining tower, resulting in the absence of interlocking shutdown systems to mitigate leakage risks

developed to improve the quality of knowledge base construction and modeling capabilities, such as HAZOPTOOL (Karvonen, 1990), (Chae et al., 1994), OptHAZOP (Khan and Abbasi, 1997a), TOPHAZOP (Khan and Abbasi, 1997b), EXPERTOP (Khan and Abbasi, 2000), and ExpHAZOP+ (Rahman et al., 2009). These systems rely on human-defined rules, which are challenging to implement due to the effort needed to formulate them and the difficulty in sourcing high-quality rules. Moreover, the existing rules typically analyze deviation scenarios in common unit operations, but fail to address causes and consequences in upstream and downstream processes, making them inadequate for HAZOP analysis in complex chemical processes.

To enhance the applicability and precision of intelligent HAZOP analysis systems, model-based expert systems have garnered significant attention from researchers. Catino and Ungar (1995) employed a library of qualitative differential equations to predict the potential system behaviors and developed a prototype system named QHI. At the same time, Vaidhyanathan and Venkatasubramanian (1995) proposed a modeling approach using directed graphs to represent causal relationships between process variables in chemical process systems, addressing the limitations of traditional methods in representing general HAZOP knowledge. Furthermore, to mitigate the issue of excessive irrelevant results generated by the system, Vaidhyanathan and Venkatasubramanian (1996) combined quantitative knowledge to filter and categorize HAZOP analysis outcomes, reducing the occurrence of unreasonable analysis results. Given the discrete event characteristics associated with intermittent operations, Srinivasan and Venkatasubramanian (1996) introduced a dual-layer Petri net modeling framework, resulting in the development of the Batch HAZOPExpert system (Srinivasan and Venkatasubramanian, 1998a, 1998b) and its subsequent enhancement through the incorporation of quantitative process information (Venkatasubramanian et al., 2000). To improve the model's expressiveness, Zhao et al. (2005) proposed a dual-layer model combining directed graphs and Petri nets, culminating in the creation of the PHASuite system. Additionally, Cui et al. (2008) introduced a hierarchical directed graph model that explicitly represents process variables, deviations, and interrelations among deviations across three dimensions, integrating it with quantitative knowledge to enhance reasoning efficiency and accuracy. To depict the hierarchical functional relationships within the process system, Rossing et al. (2010) introduced the Multilevel Flow Model (MFM), which provides a robust framework for HAZOP analysis. This hierarchical and goal-oriented approach aids in identifying and analyzing of potential hazards by tracing fault propagation through various functional levels. In contrast, Rodríguez and Mata (2012) proposed the use of D-higraphs to provide a more detailed representation of the system's dynamic behaviors and interactions, which is crucial for analyzing system performance and operational safety under varying conditions. Although the development of model libraries enhances the depth and breadth of reasoning, the modeling process is time-consuming and requires long-term verification and updates, presenting challenges for the widespread application of model-based expert systems.

In response to the challenges associated with model or rule verification in the first two generations of intelligent HAZOP analysis systems, a reasoning approach based on historical HAZOP analysis cases has been proposed. Zhao et al. (2009) utilized ontology and Case-Based Reasoning (CBR) technology to develop the PetroHAZOP software, which constructed six ontologies related to HAZOP analysis and enabled the automatic retrieval of similar HAZOP analysis cases through a case similarity algorithm. Daramola et al. (2013) introduced a method for the semi-automatic construction of domain ontologies using Natural Language Processing (NLP) technology and developed the intelligent software KROSA. Single et al. (2020) constructed a process safety knowledge ontology and implemented a risk propagation method that integrates ontology reasoning with case-based reasoning and support vector machine algorithms. At the same time, (Mao et al., 2020) constructed process safety knowledge graph based on the HAZOP analysis

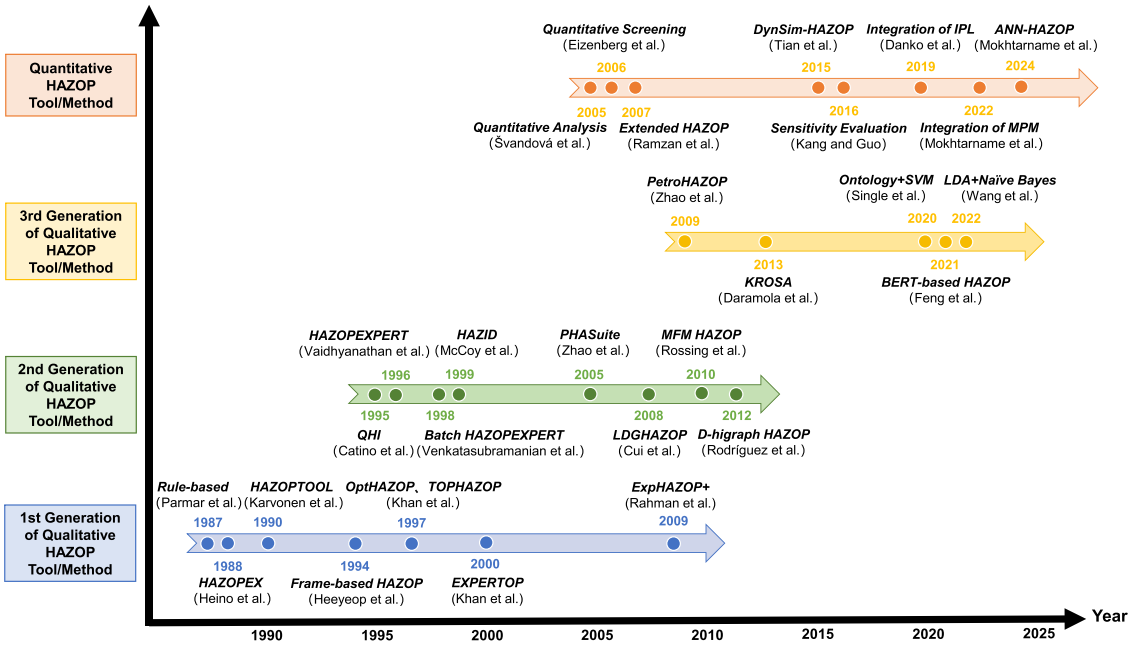


Fig. 1. The evolutionary path of representative HAZOP tools.

cases for delayed coking process, and envisaged that an automated filling of the HAZOP analysis suggestion column could be realized based on the knowledge graph. Feng et al. (2021) applied Bidirectional Encoder Representation from Transformers (BERT) to analyze HAZOP cases, achieving more precise predictions of the severity of consequences associated with risk scenarios in HAZOP analysis. Wang and Gu (2022) used a Latent Dirichlet Allocation (LDA) model to extract associations between parameter deviations, their causes, and consequences from historical cases, and employed the Naive Bayes algorithm to predict risk levels. Overall, case-based reasoning methods have proven effective by leveraging historical cases. However, as the case repository continues to grow, the managing and updating of the case database become increasingly complex, and the auxiliary capabilities of CBR technology may reach a plateau.

The intelligent HAZOP analysis systems mentioned above are all designed using qualitative reasoning approaches. To reduce dependence on expert knowledge and enhance the precision of cause-and-consequence analyses, various quantitative model-based reasoning methods have been proposed (Danko et al., 2019; Eizenberg et al., 2006; Kang and Guo, 2016; Mokhtarname et al., 2022, 2024; Ramzan et al., 2007b, 2007a; Švandová et al., 2005; Tian et al., 2015). These methods are typically implemented through dynamic simulations, which require an accurate, plant-specific process dynamic model to yield reliable analysis results (Mokhtarname et al., 2024). However, constructing process dynamic models is a complex and time-consuming task that requires reliable data, which may not always be available. Consequently, this paper concentrates on intelligent HAZOP analysis systems based on qualitative reasoning approaches.

Until now, existing intelligent HAZOP reasoning methods based on qualitative reasoning continue to encounter significant challenges, most notably their limited ability to generalize across diverse chemical processes. The rule-based and model-based reasoning methods inherently rely on custom rules or models for each process, which makes them impractical for diverse applications. In contrast, case-based reasoning methods primarily rely on pattern recognition and tend to emphasize equipment-level similarities. While this approach can be effective for pinpointing risks in isolated components, it often falls short by overlooking the broader context of process topologies and control structures that define the behavior of chemical systems. Therefore, case-based methods struggle to achieve precise matching between analogous cases, particularly when process configurations differ significantly. Consequently, the neglect of the comprehensive process structure limits the applicability of these methods to fully capture the safety risks posed by diverse process designs.

To enhance the understanding of how hazard and operability issues can arise in a process and how they can be mitigated, a next-generation of intelligent HAZOP analysis method is required to seamlessly combine factual data about the chemical processes with the textual logic from previous HAZOP analysis cases, generating actionable knowledge that provides substantial evidence for conducting reliable automated HAZOP analyses. Therefore, HAZOPCT, a tool which focuses on improving the completeness of HAZOP analysis during design phase, is proposed based on a unified ontology and graph machine learning. By utilizing a unified ontology for the knowledge graph, both the comprehensive process structures and HAZOP analysis cases are represented consistently. This consistency aids in developing a comprehensive understanding of the system's hazard and operability issues. Furthermore, compared to previous methods, graph machine learning enables a more precise identification of causal interactions and process topologies (Scholkopf et al., 2021). This approach leverages the rich structural information inherent in knowledge graphs by capturing local patterns and intricate relationships through fine-grained data representations. By accurately reflecting the underlying process structures, HAZOPCT enhances its generalization capabilities, thereby enabling more effective reasoning and accurate predictions when confronted with incomplete or missing data. Unlike current intelligent HAZOP reasoning methods that generate complete HAZOP analysis tables from scratch, HAZOPCT discovers deficiencies in existing HAZOP analysis tables and recommends the most relevant HAZOP analysis results from similar cases to complete these tables. The major contributions of this research are as follows:

- A novel tool, HAZOPCT, is proposed to address deficiencies in existing HAZOP analysis tables, thereby enhancing the comprehensiveness of HAZOP analysis.
- An automated method for constructing knowledge graphs is proposed to integrate all relevant information necessary for HAZOP analysis and previous HAZOP analysis cases, thereby streamlining the data preparation process.

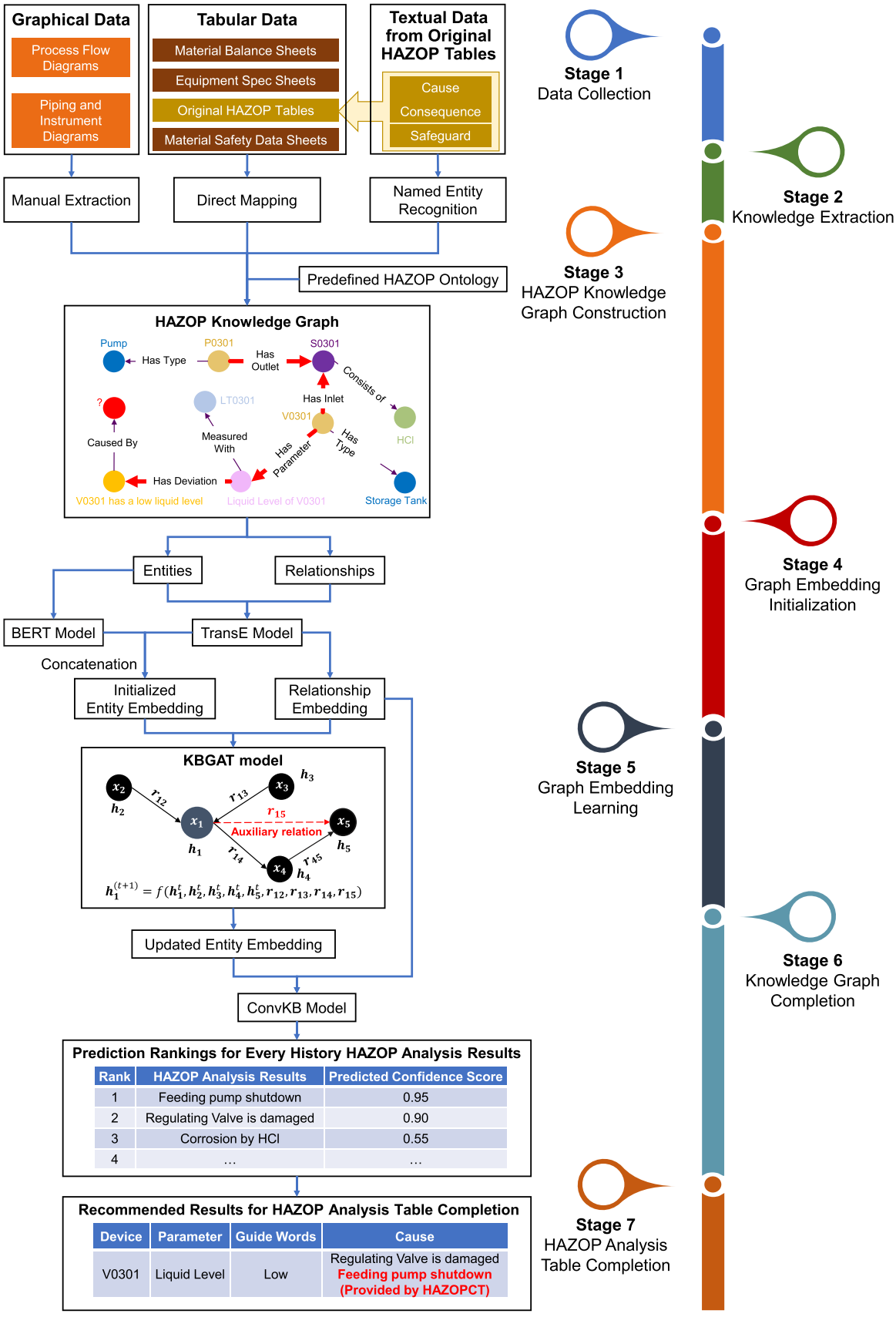


Fig. 2. The overall research framework.

Table 2
Input data specifications for HAZOPCT.

Data Source	Explanation	Detailed requirements
P&ID	An articulate drawing of a processing plan that entails the piping and process equipment with its instrumentation and control machinery	• Comply with industry standard ISO 10628 (International Organization for Standardization, 2014) • Accurately reflect the layout of all equipment, pipes, instruments, and control systems
PFD	An articulate drawing that represents the process sequence and material flow between units in the chemical process	• Comply with industry standard ISO 10628 (International Organization for Standardization, 2014) • Clearly mark the operational conditions for each process unit
Material Balance Sheet	A table that tracks material flow and balance across the system, ensuring mass conservation	• Accurately reflect all raw materials, reaction products, and by-products in the process • Ensuring conservation of mass between inputs and outputs
Equipment Specification Sheet	A table that specifies technical details for each piece of equipment in the process	• Provide specifications for each equipment that is necessary for HAZOP analysis, such as type, material, design pressure, and temperature
MSDS	Safety data for chemicals used, including their hazards, storage, and handling precautions	• Collect official MSDSs that include all relevant chemicals from material balance sheet
HAZOP Analysis Table	A table that documents the systematic analysis results of every process parameter deviation	• Comply with industry standard IEC 61882 (International Electrotechnical Commission, 2016) • Provide full risk assessment details for every possible process parameter deviation, such as causes, consequences and safeguards.

- The decision-making basis in HAZOP reasoning is elucidated through an attention weight network, thereby enhancing the interpretability of the model's predictions.

2. Methodology

2.1. Research framework overview

The overall research framework is illustrated in Fig. 2, which shows the workflow from data collection through knowledge graph construction, and graph embedding learning, culminating in the completion of the HAZOP analysis table. The methodology is divided into the following seven key stages:

Stage 1: data collection

We began by collecting preliminary design documents from multiple chemical plants. For each plant, six types of heterogeneous data essential for HAZOP analysis have been prepared: Piping and Instrumentation Diagrams (P&IDs), Process Flow Diagrams (PFDs), material balance sheets, equipment specification sheets, Material Safety Data Sheets (MSDSs), and HAZOP analysis tables. Table 2 provides a description for the six types of data sources, and the requirement for different data sources is also specified.

Stage 2: knowledge extraction

Then, an automated method is employed to extract structured information from these data sources. This method enables HAZOPCT to identify key entities, such as equipment, streams, chemicals, process parameters, and safety measures, along with their relationships needed for the HAZOP analysis.

Stage 3: HAZOP knowledge graph construction

Afterward, a HAZOP knowledge graph is constructed based on a predefined ontology. This ontology semantically integrates the various data sources, enabling HAZOPCT to represent complex relationships and perform reasoning across them. The HAZOP knowledge graph is formally represented as $G = (V, E, \varepsilon)$, where:

- $V = \{v_1, v_2, \ldots, v_N\}$ denotes the set of entities, with N representing the total number of entities.
- $E = \{e_{ij}\}$ represents the set of relationships, where each relationship e_{ij} denotes the relationship between entities v_i and v_j.
- ε contains all facts represented in the graph, expressed as triplets (h, r, t), where $h, t \in V$ and $r \in E$. These triplets denote the relationships between entities. For example, to express that a low liquid level in storage tank V0301 is caused by the shutdown of the upstream feed pump P0301 due to a malfunction, the corresponding triplet would be:

(*V0301 has a low liquid level*, *caused by*, *P0301 shutdown due to malfunction*)

Stage 4: graph embedding initialization

The next stage involves initializing the graph embedding, where we convert the knowledge graph into a high-dimensional vector space. This process allows for the numerical representation of the graph's structure and entity attributes, making it suitable for further computational analysis.

Stage 5: graph embedding learning

Once initialized, the entity embeddings are optimized through a graph attention network that considers the significance of each neighboring entity in relation to the central entity. This approach enables the model to concentrate on the most relevant aspects of the graph, thereby facilitating the learning of more meaningful features.

Stage 6: knowledge graph completion

Based on the optimized embeddings, a convolutional neural network is utilized to evaluate the confidence scores for each possible HAZOP analysis results. These results are then ranked in descending order according to their confidence scores.

Stage 7: HAZOP analysis table completion

Finally, potentially overlooked results are identified according to the prediction rankings. For example, if a deviation in a process parameter is neglected in a HAZOP analysis table, and the prediction rankings indicate that a specific equipment malfunction is most likely to cause the deviation, HAZOPCT can recommend adding this causal relationship to the table. This approach enhances the efficiency of identifying deficiencies within the HAZOP analysis tables.

Table 3
Commonly used notations.

Notations	Descriptions
G	Knowledge graph
V	Set of entities in the knowledge graph
E	Set of relationships in the knowledge graph
ε	Set of triplets in the knowledge graph
h	Head entity
r	Relationship
t	Tail entity
N	Number of entities in the knowledge graph
$\vec{h}_i$	Initial vector representation of entity h_i
$\vec{h}'_i$	Output vector representation of entity h_i generated by the graph attention network
$(\vec{v}_h, \vec{v}_r, \vec{v}_t)$	Final vector representation of the triplet (h, r, t)
α_{ij}	Relative attention weight assigned to the neighboring entity h_j of entity h_i
c_{ij}	Vector representation of the neighboring entity h_j of entity h_i
$S(h_i)$	Set of neighbors and associated relationships of entity h_i

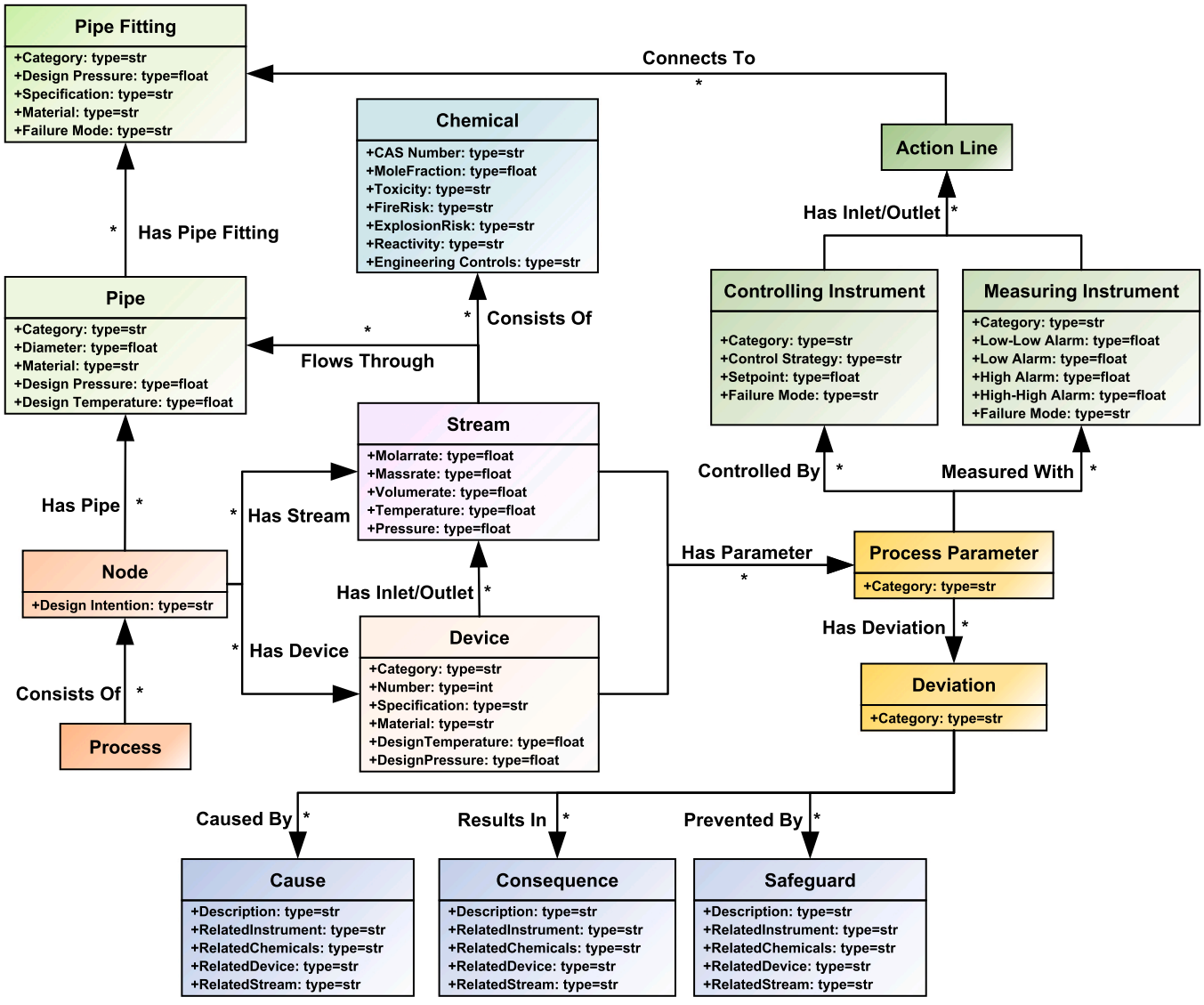


Fig. 3. An ontology model of the HAZOP knowledge graph.

Commonly used notations are detailed in Table 3, unless specified otherwise.

2.2. Construction of the HAZOP knowledge graph

2.2.1. Design of the HAZOP ontology

This study presents a HAZOP ontology designed to model concepts and relationships pertinent to HAZOP analysis, thereby facilitating the integration of multi-source process safety information and the construction of the HAZOP knowledge graph. The HAZOP ontology contains concepts related to devices, streams, pipework and fittings, chemicals, controlling and measuring instrument, as well as HAZOP analysis for the process.

The ontology is based on specific concept definitions derived from the OntoCAPE ontology (Morbach et al., 2007) and the ontology from PetroHAZOP (Zhao et al., 2009). In contrast to the two existing ontologies, the ontology proposed herein integrates control and pipeline information relevant to process design by introducing five key concepts: "*Control_Instrument*", "*Measuring_Instrument*", "*Action_Line*", "*Pipe*", and "*Pipe_Fitting*". Additionally, it distinctly conceptualizes material flow within the system, emphasizing chemical composition and properties as critical attributes. These enhancements enable more precise identification of deviations in process parameters and improve the localization of instruments. For instance, the modified ontology can effectively differentiate between the overhead and bottom product streams of a distillation column, facilitate accurate associations with temperature indicators at these specific points, and enhance the granularity of the HAZOP knowledge graph. The HAZOP ontology is illustrated in Fig. 3.

2.2.2. Integration of various data into a HAZOP knowledge graph

Building upon the defined HAZOP ontology, this section outlines the specific methodologies employed to construct the HAZOP knowledge graph from various data sources. Tailored information extraction techniques were developed for each data type:

- **Graphical data (PFDs and P&IDs):** We manually extracted process topologies and control structures for each process design, which include streams, equipment, instruments, valves, their identification numbers, and the relationships between these components.
- **Tabular data (equipment specification sheets, MSDSs, material balance sheets, and HAZOP analysis tables):** We directly mapped these tables to graph structures. The equipment and streams in P&IDs are aligned with the corresponding equipment specification sheet, HAZOP analysis table, or material balance sheet through their identification numbers. The chemical compositions of the streams are obtained from the material balance sheets, and subsequently

Table 4
A Framework for HAZOP Knowledge Graph Construction.

INPUT: *P&IDs*, *PFDs*, *HAZOP analysis tables*, *Material balance sheets*, *Equipment specification sheets*, *MSDSs, Process IDs* (Tag_{pro})
Initialize HAZOP knowledge graph as an empty set {}
Extract HAZOP nodes $N = \{n_1, n_2, \ldots\}$ from the HAZOP analysis tables
For each node n_i in N, do:
Add the triplet (Tag_{pro}, consists of, n_i) to the HAZOP knowledge graph
Select corresponding P&IDs for the designated node n_i
Select corresponding PFDs for the designated node n_i
Extract the equipment $E = \{e_1, e_2, \ldots\}$ in P&IDs
For each equipment e_j in E, do:
Extract the equipment ID Tag_{equip} from P&IDs
Select the corresponding equipment specification sheet
Extract the properties $K = \{k_1, k_2, \ldots, k_M\}$ and their corresponding values $V = \{v_1, v_2, \ldots, v_M\}$
For each property-value pair (k_l, v_l), do:
Add the triplet (Tag_{equip}, k_l, v_l) to the HAZOP knowledge graph
end for
Extract the inlet and outlet stream IDs Tag_{stream} from PFDs
Add the triplet (Tag_{equip}, *has inlet/outlet*, Tag_{stream}) to the HAZOP knowledge graph
Select the corresponding material balance sheet
Extract the chemicals $CHE = \{c_1, c_2, \ldots\}$ from the sheet
For each chemical c_k in CHE, do:
Add triplet ($Tag_{stream,}$ consists of, c_k) to the HAZOP knowledge graph
Select the corresponding MSDS for c_k
Extract the properties $P = \{p_1, p_2, \ldots, p_T\}$ and the values $R = \{r_1, r_2, \ldots, r_T\}$ from the MSDS
For each property-value pair (p_q, r_q), do:
Add triplet (c_k, p_q, r_q) to the HAZOP knowledge graph
end for
end for
end for
Select the corresponding HAZOP analysis table
Extract all the possible deviations $DEV = \{d_1, d_2, \ldots\}$ in the table
For deviation d_m in DEV, do:
Extract the equipment ID Tag_{equip} or the stream ID Tag_{stream} from the HAZOP analysis table
Extract the description of the deviation *DEVIATION* from the HAZOP analysis table
Add triplet (Tag_{equip} or Tag_{stream}, *hasdeviation*, *DEVIATION*) to the HAZOP knowledge graph
Extract the parameter *PARAM* from the HAZOP analysis table
Add triplet (*DEVIATION*, relatedparameter, *PARAM*) to the HAZOP knowledge graph
Extract all the controlling instruments $C_I = \{ci_1, ci_2, \ldots\}$ corresponding to *PARAM* from P&IDs
For controlling instrument ci_n in C_I, do:
Add triplet (*PARAM*, controlledby, ci_n) to the HAZOP knowledge graph
Extract all the measuring instruments $M_I = \{mi_1, mi_2, \ldots\}$ corresponding to *PARAM* from P&IDs
For measuring instrument mi_p in M_I, do:
Add triplet (*PARAM*, measuredwith, mi_p) to the HAZOP knowledge graph
Extract the guide word *GUIDE* from the HAZOP analysis table
Add triplet (*DEVIATION*, hastype, *GUIDE*) to the HAZOP knowledge graph
Extract the possible cause *CAUSE* to the deviation from the HAZOP analysis table
Add triplet (*DEVIATION*, causedby, *CAUSE*) to the HAZOP knowledge graph
Extract the related instrumentation, chemicals, devices or stream *REF* from *CAUSE*
Add triplet (*CAUSE*, relatedto, *REF*) to the HAZOP knowledge graph
Extract the consequence *CONSEQUENCE* to the deviation from the HAZOP analysis table
Add triplet (*DEVIATION*, resultsin, *CONSEQUENCE*) to the HAZOP knowledge graph
Extract the relevant instrumentation, chemicals, devices, or stream *REF* from *CONSEQUENCE*
Add triplet (*CONSEQUENCE*, relatedto, *REF*) to the HAZOP knowledge graph
Extract the prevention *PREVENTION* to the deviation from the HAZOP analysis table
Add triplet (*DEVIATION*, preventedby, *PREVENTION*) to the HAZOP knowledge graph
Extract the relevant instrumentation, chemicals, devices, or stream *REF* from *PREVENTION*
Add triplet (*PREVENTION*, relatedto, *REF*) to the HAZOP knowledge graph
end for
end for
Store the constructed HAZOP knowledge graph in a Neo4j graph database for further analysis and querying

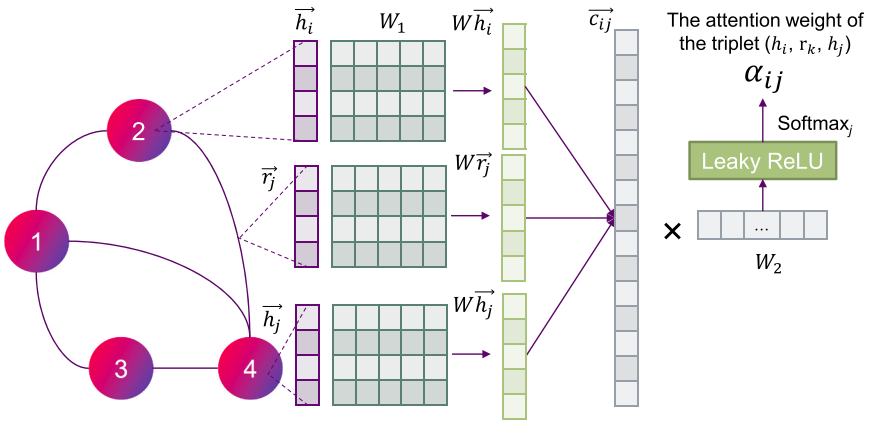


Fig. 4. The attention weight calculation method of KBGAT.

aligned with the MSDS data using a synonym table to address potential discrepancies in chemical nomenclature.

- **Textual data:** We utilized the BERT-CRF framework (Souza et al., 2020) to extract chemical and equipment entities, particularly identifiers from descriptions of causes, consequences, and safeguards in HAZOP analysis tables. The extracted entities were then added to the HAZOP knowledge graph. For instance, a HAZOP analysis table identified that the low feed flow rate of heat exchanger E0303 might be caused by a leakage in feed heat exchanger E0302. We identified the named entity *"E0302"* from the description of cause and established a connection in the knowledge graph between the cause analysis entity and the equipment entity E0302. The named entity recognition method enhances HAZOPCT by enabling more precise identification of equipment associated with root causes:

(*leakage in feed heat exchanger E0302*, *related to*, *E0302*)

Unfortunately, we were unable to collect detailed design information about pipes and pipe fittings for some of the design works. As a result, entities for pipes and pipe fittings were not included in the HAZOP knowledge graph. Table 4 outlines the detailed framework for integrating various data into a HAZOP knowledge graph.

2.3. HAZOPCT: a tool based on knowledge graph reasoning

2.3.1. Initializing graph embeddings by integrating structure and semantics

To improve computational efficiency and provide semantic support for downstream tasks such as link prediction, this study represents entities and relationships as low-dimensional, dense vectors. To effectively capture both structural and semantic information within the HAZOP knowledge graph, we proposed a method that integrates these two aspects into the vector representations of entities and relationships. These integrated vectors serve as the initial input for the graph attention network.

To obtain structural information, the TransE model (Bordes et al., 2013) is used to generate structural embeddings for each entity and relationship. The structural embeddings for entities are denoted as $\boldsymbol{H_{str}} = \{h_1, h_2, \cdots, h_N\}$, and those for relationships are denoted as $\boldsymbol{R_{str}} = \{r_1, r_2, \cdots, r_Q\}$. To capture the semantic information associated with each entity, the BERT pre-trained language model (Devlin et al., 2019) is utilized to generate semantic embeddings from the textual entity descriptions, which are denoted as $\boldsymbol{H_{sem}} = \{h_1', h_2', \cdots, h_N'\}$. Then the structural and semantic embeddings for entities are concatenated horizontally to form integrated vectors $\boldsymbol{H}$, which serve as the initialized entity embeddings for further processing.

$$\boldsymbol{H} = \boldsymbol{H_{str}} \parallel \boldsymbol{H_{sem}} \tag{1}$$

$$\boldsymbol{R} = \boldsymbol{R_{str}} \tag{2}$$

2.3.2. Learning graph embeddings using graph attention networks

This study employs the Knowledge Base Graph Attention Network (KBGAT) (Nathani et al., 2019) to optimize knowledge graph embeddings. Graph Attention Networks (GATs) utilize a self-attention mechanism to emphasize the relationships between each entity and its neighboring entities. The attention weights, which signify the importance of each relationship, are learned through the algorithm, enabling the model to update the vector representation of each entity accordingly. Due to their strong interpretability, GATs have been widely adopted in knowledge graph embedding tasks. The methodology for calculating attention weights is illustrated in Fig. 4.

The importance of features from a neighboring entity t_j and the corresponding relationship r_j relative to the embedding of central entity h_i is quantified as follows:

$$\vec{c}_{ij} = \mathbf{W_1}\left[\vec{h}_i \parallel \vec{t}_j \parallel \vec{r}_j\right] \tag{3}$$

$$\alpha_{ij} = softmax_j\left(LeakyReLU\left(\boldsymbol{W_2}\,\vec{c}_{ij}\right)\right) \tag{4}$$

Here, α_{ij} denotes the attention weight for the triplet (h_i, r_j, t_j), $\boldsymbol{W_1}$ and $\boldsymbol{W_2}$ are trainable parameters. Initially, the vector representations of the head entity $\vec{h}_i$, relationship $\vec{r}_j$, and tail entity $\vec{t}_j$, which are derived from the previously described initialization method, are concatenated horizontally. The attention weight for each triplet is then computed by linear transformations, followed by LeakyReLU activation and a softmax layer to obtain normalized attention weights.

In addition to capturing the importance of direct neighbors, KBGAT extends its approach to include second-order neighbors of the central entity. Since no direct relationship connects second-order neighbors to the central entity, an auxiliary relationship $r_{k,aux}$ is introduced. Its vector is initialized as the sum of vectors corresponding to relationships along the path. Additionally, a multi-head self-attention mechanism is employed to update the vector representation of the central node:

$$\vec{h}_i' = \Big\Vert_{m=1}^{M} \boldsymbol{W_3}\left(\sum_{j\in N_i}\alpha_{ij}^m \vec{c}_{ij}^m\right) \tag{5}$$

Here, M represents the number of attention heads, and N_i denotes the set of first-order neighbors and second-order neighbors of entity h_i. Each attention head m computes its own set of attention weights α_{ij}^m and feature representations $\vec{c}_{ij}^m$.

Entity embeddings are updated within a single graph attention layer as described above. Since the KBGAT model comprises multiple graph attention layers, each subsequent layer refines the entity embeddings by incorporating information from the preceding layer, thereby facilitating the gradual accumulation of more expressive features. In the final layer

Table 5
A framework for the graph attention network algorithm.

INPUT: HAZOP knowledge graph $\boldsymbol{G}$= {(h_1, r_1, t_1), (h_2, r_2, t_2), …, (h_T, r_T, t_T)}, Num of attention heads N, Entity set H and relationship set R, Input dimension K, Output dimension M, Initialized entity embeddings $\boldsymbol{H}$, Initialized relationship embeddings $\boldsymbol{R}$, Training Epochs L, Margin δ, Ratio of negative to positive triplets T

For entity h_i in H, do:
  Retrieve the corresponding initial vector $\boldsymbol{h}_{i,init}$ in $\boldsymbol{H}$
  Set $\boldsymbol{h}_i(0) = \boldsymbol{h}_{i,init}$
  Find the set of first-order and second-order neighbors with the corresponding relationships $S(h_i)$={(t_j, r_j)}
end for
Initialize $\mathbf{W}_{\mathbf{mid}}$ of size $K \times (N*K)$ with all elements as 1
Initialize $\mathbf{W}_{\mathbf{final}}$ of size $M \times (N*K)$ with all elements as 1
Initialize $\mathbf{W}_{\mathbf{trans}}$ of size $M \times K$ with all elements as 1
For each attention head m=1 to N do ← Initialize the parameters for multi-head attentions
  Initialize $\mathbf{W}_{1,m}$ of size $K \times 3K$ with all elements as 1
  Initialize $\mathbf{W}_{2,m}$ of size $1 \times K$ with all elements as 1
end for
For each epoch e=1 to L, do:
  For each layer l=1 to E-1, do:
    For each entity h_i in H, do:
      For each attention head m=1 to N, do:
        Initialize $\boldsymbol{h}_{i,m}$ of size $K \times 1$ with all elements as 0
        For each neighbor (t_j, r_j) in $S(h_i)$, do:
          $\boldsymbol{C}_{ij,m}=\mathbf{W}_{1,m} \bullet [\boldsymbol{h}_i(l\text{-}1) \| \boldsymbol{t}_j(l\text{-}1) \| \boldsymbol{r}_j]$
          $z_{ij,m}$= LeakyReLU ($\mathbf{W}_{2,m} \bullet \boldsymbol{C}_{ij,m}$) ←Calculate the absolute attention weight
          $\sigma_{ij,m}$= exp($z_{ij,m}$) / Σ_k exp($z_{ik,m}$) ←Calculate the normalized attention weight
          $\boldsymbol{h}_{i,m}= \boldsymbol{h}_{i,m}+\sigma_{ij,m}\, \boldsymbol{C}_{ij,m}$
        end for
      end for
      $\boldsymbol{h}_i(l)= \mathbf{W}_{\mathbf{mid}}\, [\boldsymbol{h}_{i,1} \| \boldsymbol{h}_{i,2} \| \ldots \| \boldsymbol{h}_{i,N}]$
    end for
  end for
  For entity h_i in H, do:
    Initialize $\boldsymbol{h}_i(E)$ of size $K \times 1$ with all elements as 0
    For each attention head m=1 to N, do:
      For each neighbor (t_j, r_j) in $S(h_i)$, do:
        $\boldsymbol{C}_{ij,m}=\mathbf{W}_{1,m} \bullet [\boldsymbol{h}_i(E\text{-}1) \| \boldsymbol{t}_j(E\text{-}1) \| \boldsymbol{r}_j]$
        $z_{ij,m}$= LeakyReLU ($\mathbf{W}_{2,m} \bullet \boldsymbol{C}_{ij,m}$) ←Calculate the absolute attention weight
        $\sigma_{ij,m}$= exp($z_{ij,m}$) / Σ_k exp($z_{ik,m}$) ←Calculate the normalized attention weight
        $\boldsymbol{h}_i(E)= \boldsymbol{h}_i(E)+\sigma_{ij,m}\, \boldsymbol{C}_{ij,m}$
      end for
    end for
    $\boldsymbol{h}_i(E)= \mathbf{W}_{\mathbf{final}} \bullet \boldsymbol{h}_i(E)$ / N
  end for
  $\boldsymbol{h}_i(final) = \mathbf{W}_{\mathbf{trans}} \bullet \boldsymbol{h}_i(0) + \boldsymbol{h}_i(E)$
  Initialize totalloss L as 0
  For each positive triplet (h_i, r_i, t_i) in $\boldsymbol{G}$, do:
    Pos_norm = $\| \boldsymbol{h}_i(final) + \boldsymbol{r}_i - \boldsymbol{t}_i(final) \|_1$ ←Calculate the positive score
    For time=1 to T, do:
      Generate a negative triplet (h_i, r_i, t_k) ∉ $\boldsymbol{G}$
      Neg_norm = $\| \boldsymbol{h}_i(final) + \boldsymbol{r}_i - \boldsymbol{t}_k(final) \|_1$ ←Calculate the negative score
      rankloss = max (Neg_norm - Pos_norm + δ, 0)
      L = L + rankloss
    end for
  end for
  Update parameters $\mathbf{W}_{\mathbf{mid}}$, $\mathbf{W}_{\mathbf{final}}$, $\mathbf{W}_{\mathbf{trans}}$, {$\mathbf{W}_{1,m}$}, {$\mathbf{W}_{2,m}$} using gradients computed from L
end for
Return optimized entity embeddings $\boldsymbol{H}_{\mathbf{opt}}$={$\boldsymbol{h}_i(final)$} and relationship embeddings $\boldsymbol{R}$

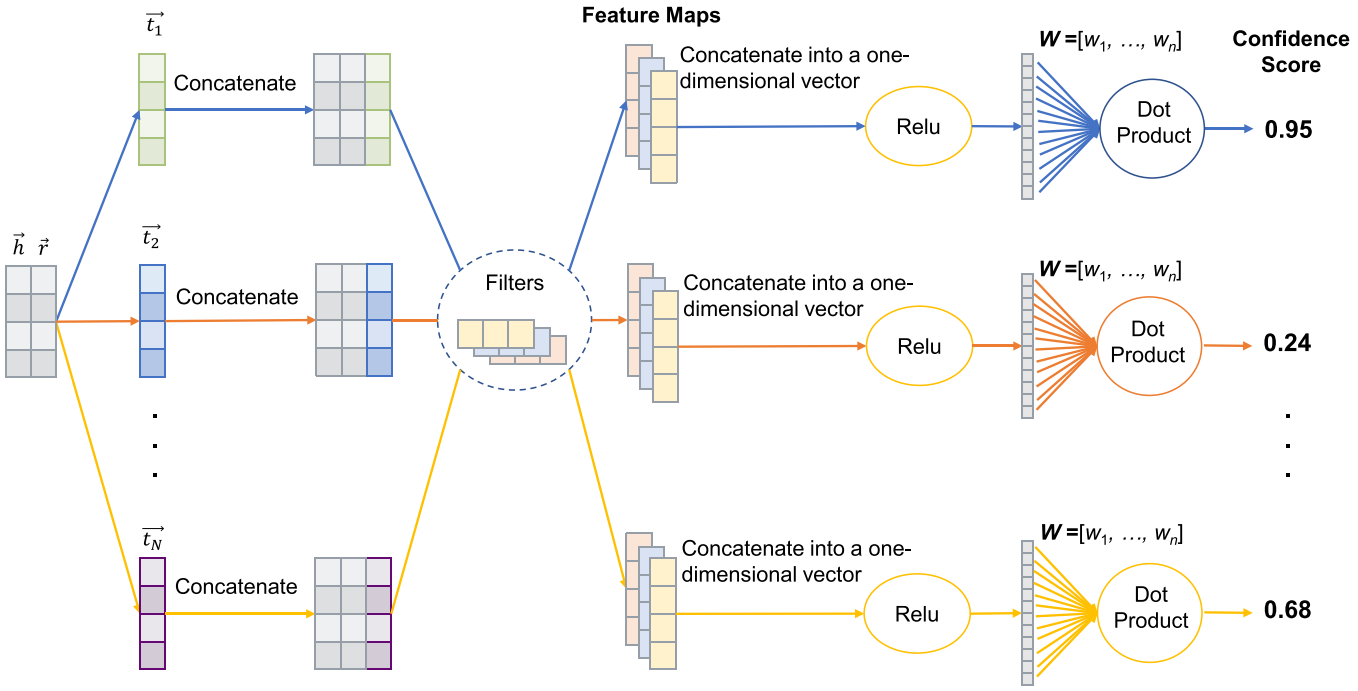


Fig. 5. The basic architecture of the ConvKB model.

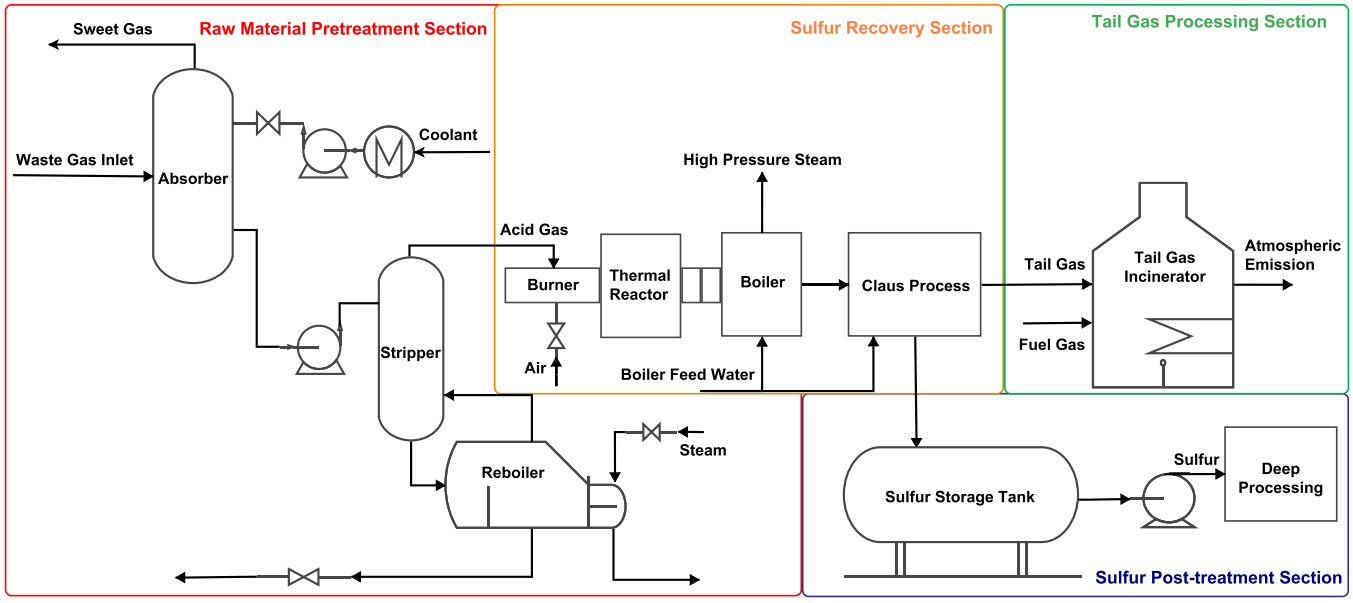


Fig. 6. A schematic diagram of sulfur recovery process.

of the model, instead of concatenating the vector representations from the various attention heads, we compute the average to derive the final updated entity embedding:

$$\vec{h}'_i = \frac{1}{M}\boldsymbol{W}_4\left(\sum_{m=1}^{M}\sum_{j\in N_i}\alpha_{ij}^m \vec{c}_{ij}^m\right) \tag{6}$$

To prevent the loss of initial embedding information during the entity embedding update process, a linear mapping layer projects the initial entity embedding $\vec{h}_i$ into the KBGAT output space. This mapped vector is then added to $\vec{h}'_i$, yielding the final embedding for each triplet (h_i, r_k, h_j), denoted as $(\vec{v}_h; \vec{v}_r; \vec{v}_t)$. Table 5 presents a framework for the graph attention network algorithm.

2.3.3. *Completing the knowledge graph with convolutional neural networks*

After obtaining vector embeddings for each triplet, this study utilizes ConvKB model (Nguyen et al., 2018) as a decoder to generate confidence scores. Fig. 5 illustrates the architecture of ConvKB. Initially, the k-dimensional embeddings of the head entity $\vec{v}_h$, the relationship $\vec{v}_r$, and the tail entity $\vec{v}_t$, obtained through KBGAT, are vertically concatenated into a $k \times 3$ matrix $\boldsymbol{A}$:

$$\boldsymbol{A} = \left[\vec{v}_h; \vec{v}_r; \vec{v}_t\right] \tag{7}$$

In the convolutional layer, different convolutional filters $\omega_i \in R^{1\times 3}$ $(i = 1,2,\ldots,\Omega)$ are iteratively applied to each row of matrix $\boldsymbol{A}$, producing k-dimensional feature maps. These feature maps are activated using the

Table 6
Description of the 19 Design Works Related to Sulfur Recovery Process.

Design Work Number	Main Components of Waste Gases	Main Product	Types of Waste Gas Absorbents	Type of Catalyst	Sulfur Recovery Scale (kt/y)
Design Work 1	H_2S (42.9 kg%), CO_2 (52.2 kg%)	CS_2	$C_4H_6O_3$	CoO, MoO_3, Al_2O_3	73.84
Design Work 2	N_2 (72.0 kg%), CO_2 (14.1 kg%), SO_2 (0.5 kg %)	CO_2(Purified), S	$NH_3{\cdot}H_2O$	TiO_2, MoO_3, Al_2O_3	166.59
Design Work 3	N_2 (65.8 kg%), CO_2 (27.7 kg%), SO_2 (0.4 kg %)	S	Hollow fiber membrane	Cu-Al_2O_3	68.89
Design Work 4	N_2 (74.7 kg%), O_2 (21.9 kg%), SO_2 (0.3 kg %)	C_4H_8S	CH_3OH	Al_2O_3	0.86
Design Work 5	N_2 (71.2 kg%), CO_2 (18.5 kg%), SO_2 (0.2 kg %)	H_2SO_4, Na_2SO_4	NaOH	V_2O_5, K_2O, SiO_2	8.50
Design Work 6	N_2 (66.8 kg%), CO_2 (16.9 kg%), SO_2 (0.5 kg %)	Na_2SO_4, CO_2(Purified)	NH_4HSO_3	None	10.07
Design Work 7	N_2 (64.5 kg%), O_2 (15.5 kg%), SO_2 (0.3 kg %)	$(NH_4)_2SO_4$	Na_2HPO_4	None	9.04
Design Work 8	N_2 (72.0 kg%), O_2 (18.1 kg%), SO_2 (0.5 kg %)	H_2SO_4, SO_2(Liquid)	CH_3OH	Na_2O, Al_2O_3	2.63
Design Work 9	H_2S (40.4 kg%), CO_2 (51.0 kg%)	CO_2(Purified), S	Na_3PO_4, $NaHSO_3$	TiO_2, MoO_3, Al_2O_3	229.70
Design Work 10	N_2 (69.3 kg%), CO_2 (14.6 kg%), SO_2 (0.7 kg %)	S, SO_2(Liquid)	$NH_3{\cdot}H_2O$	KI, Al_2O_3	103.26
Design Work 11	N_2 (72.0 kg%), O_2 (13.3 kg%), SO_2 (0.3 kg %)	H_2SO_4	$C_5H_{13}NO_2$	TiO_2, MoO_3, Al_2O_3	137.41
Design Work 12	CH_4 (44.1 kg%), C_2H_6 (15.8 kg%), H_2S (6.1 kg%)	S	$NH_3{\cdot}H_2O$	None	11.70
Design Work 13	N_2 (74.9 kg%), CO_2 (18.1 kg%), SO_2 (0.2 kg %)	$(NH_4)_2S_2O_8$, CO_2(Liquid)	$NH_3{\cdot}H_2O$	None	15.37
Design Work 14	N_2 (75.9 kg%), CO_2 (9.8 kg%), SO_2 (0.3 kg %)	$(NH_4)_2SO_4$	Na_2S	Fe_2O_3	3.70
Design Work 15	N_2 (74.6 kg%), CO_2 (17.0 kg%), SO_2 (0.3 kg %)	CO_2(Purified), S	NaOH	None	13.70
Design Work 16	N_2 (73.4 kg%), CO_2 (13.6 kg%), SO_2 (0.2 kg %)	H_2SO_4	$NH_3{\cdot}H_2O$	V_2O_5, K_2O, SiO_2	42.20
Design Work 17	CH_4 (12.5 kg%), C_2H_6 (24 kg%), H_2S (43.4 kg%)	C_2H_6OS	$C_5H_{13}NO_2$	MoO_3, CeO_2, Al_2O_3	14.26
Design Work 18	H_2S (46.5 kg%), CO_2 (52.5 kg%)	S	$C_5H_{13}NO_2$	CoO, MoO_3, Al_2O_3	29.08
Design Work 19	N_2 (71.2 kg%), CO_2 (17.9 kg%), SO_2 (0.5 kg %)	H_2SO_3	Na_2SO_3	None	38.14

Table 7
Synonym table for aligning chemical entities.

Chemical Formal Name (English Full Name)	Synonyms
Hydrogen sulfide	H_2S; dihydrogen monosulfide; sulfane; sulfur hydride; hydrosulfuric acid (when dissolved in water); stink damp; sewer gas; sour gas
Sulfur	S; brimstone; sulfur element; flowers of sulfur; sublimed sulfur; sulfur powder
Carbon disulfide	CS_2; carbon bisulfide; dithiocarbonic anhydride
Methane	CH_4; marsh gas; methyl hydride; firedamp
Sulfur dioxide	SO_2; sulfurous anhydride; sulfurous oxide
Phosgene	$COCl_2$; carbonyl chloride; carbon oxychloride; carbonic dichloride; chloroformyl chloride; CG (military designation)
Carbon dioxide	CO_2; carbonic anhydride; carbonic acid gas; dry ice (solid form)
Oxygen	O_2; dioxygen; molecular oxygen; oxygen gas
Hydrogen	H_2; dihydrogen; molecular hydrogen; hydrogen gas
Carbon monoxide	CO; carbon (II) oxide; carbonous oxide
Carbonyl sulfide	COS; carbon oxysulfide; carbon oxide sulfide
Sulfur trioxide	SO_3; sulfuric oxide; sulfur (VI) oxide
Nitrogen	N_2; dinitrogen; molecular nitrogen; nitrogen gas
Ammonia	NH_3; azane; hydrogen nitride; ammonia gas; anhydrous ammonia
Sodium sulfate	Na_2SO_4; disodium sulfate; sulfuric acid disodium salt; Glauber's salt (decahydrate form)
Nitric oxide	NO; nitrogen monoxide
Nitrogen dioxide	NO_2; nitrogen (IV) oxide; nitrogen peroxide
Methyl diethanolamine	MDEA; N-methyl diethanolamine; 2,2′-methyliminodiethanol; methyl-di(2-hydroxyethyl)amine

ReLU function and subsequently concatenated into a single feature vector. Finally, the confidence score for each triplet is calculated through a dot product operation:

$$Conf_{\boldsymbol{A}} = \boldsymbol{W_4} \cdot \left(\Big\Vert_{m=1}^{\Omega} ReLU(\boldsymbol{A} * \boldsymbol{k^m}) \right) \tag{8}$$

Here, $\boldsymbol{k^m}$ represents the mth convolutional filter, Ω is a hyper-parameter denoting number of filters used, $*$ represents the convolution operation between matrix $\boldsymbol{A}$ and the convolution kernels, $\boldsymbol{W_4}$ represents the trainable weights used to compute the final score of the triplet, and $Conf_{\boldsymbol{A}}$ represents the confidence score corresponding to the triplet (h_i, r_k, h_j).

2.3.4. HAZOP analysis table completion using HAZOPCT

Leveraging HAZOPCT, we presented an intelligent approach for completing HAZOP analysis tables. This method is divided into two primary phases: training and inference. Prior to the training stage, we created several invalid triplets from each triplet in the training set. Specifically, we substituted the tail entity with entities that do not form a valid relationship within the knowledge graph. This process is to ensure that HAZOPCT can not only recognize valid entity relationships but also effectively distinguish invalid ones. Moreover, the embeddings for entities and relationships are initialized using the TransE model (Bordes et al., 2013) and the BERT pre-trained language model (Devlin et al., 2019).

The training phase involves a two-step procedure. First, KBGAT is trained using margin ranking loss to encode information about graph entities from their neighboring entities. In this phase, the objective is to

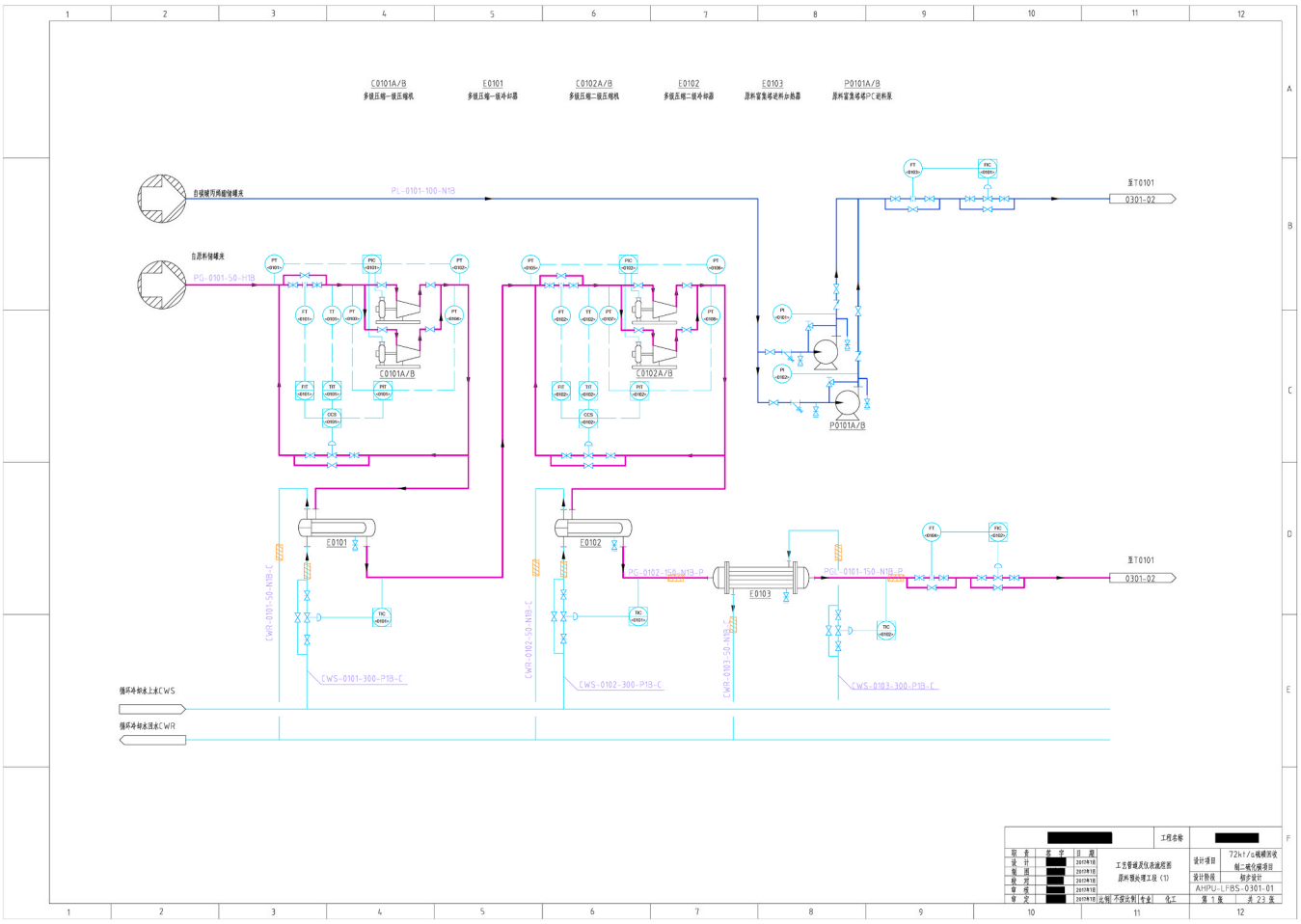


Fig. 7. An Example of the piping and instrument diagram for sulfur recovery process.

distinguish valid triples from invalid ones by employing the L_1-norm dissimilarity measure. Subsequently, the ConvKB model is trained using soft margin loss to perform knowledge graph completion. This stage focuses on ranking valid triples above invalid ones.

During the inference phase, the goal is to predict potential causes, consequences, or safeguards given a specific deviation in process parameters. Specifically, we aim to predict a triplet (h, r, t) where the tail entity t is missing. The relationship r must be chosen from one of three predefined categories: "*caused by*", "*results in*", or "*prevented by*". We constructed a set of triples by combining the head entity h, the relationship r, and every possible cause, consequence, or safeguard present in the HAZOP knowledge graph, which represents the tail entity t. The corresponding embeddings for the head entity, relationship, and tail entity are vertically concatenated and subsequently input into the ConvKB model for inference. The ConvKB model outputs the confidence scores for each triplet. Finally, all triplets are ranked in descending order based on their confidence scores, thereby generating prediction rankings, which are ordered lists of predicted triplets produced by HAZOPCT. These rankings provide recommendations for the most valuable historical analysis results.

3. Case study

3.1. Dataset construction process: from raw data to knowledge graph

3.1.1. Raw safety information for sulfur recovery process

Sulfur recovery process is a vital process for recovering sulfur from industrial waste gases, primarily utilizing hydrogen sulfide (H_2S) and sulfur dioxide (SO_2) as raw materials. At elevated temperatures (100–200°C), H_2S reacts with SO_2 to produce elemental sulfur and water. Additionally, SO_2 can be further processed into sulfuric acid (H_2SO_4), which can neutralize with bases to form sulfates. Organic byproducts, such as carbon disulfide (CS_2), tetrahydrothiophene (C_4H_8S), and ethyl sulfide (C_2H_6OS), may also form under specific conditions. Fig. 6 illustrates the schematic of sulfur recovery process, which typically consists of the following four sections.

- **Raw Material Pretreatment Section:** Preparing the feedstock for sulfur recovery process, typically sour gas. It involves the removal of contaminants such as water, carbon dioxide, and other impurities to ensure that the gas is in an optimal condition for further processing. Common operations in this section include acid gas sweetening (e.g., using ammonia or amine-based solvents) and dehydration.
- **Sulfur Recovery Section:** This is the core section of sulfur recovery process, where H_2S is converted into elemental sulfur. The most common method used here is the Claus process, which involves the reaction of H_2S with oxygen in the presence of a catalyst to produce sulfur and water. The process typically consists of two main stages: the thermal stage and the catalytic stage. In the thermal stage, part of the H_2S is combusted with oxygen to form SO_2, and in the catalytic stage, the SO_2 reacts with the remaining H_2S to produce sulfur.
- **Sulfur Post-Treatment Section:** This section is responsible for utilizing the recovered sulfur in downstream products, ensuring that it meets the required specifications for sale or storage.
- **Tail Gas Processing Section:** After sulfur recovery, the remaining gases (tail gases) contain trace amounts of sulfur compounds,

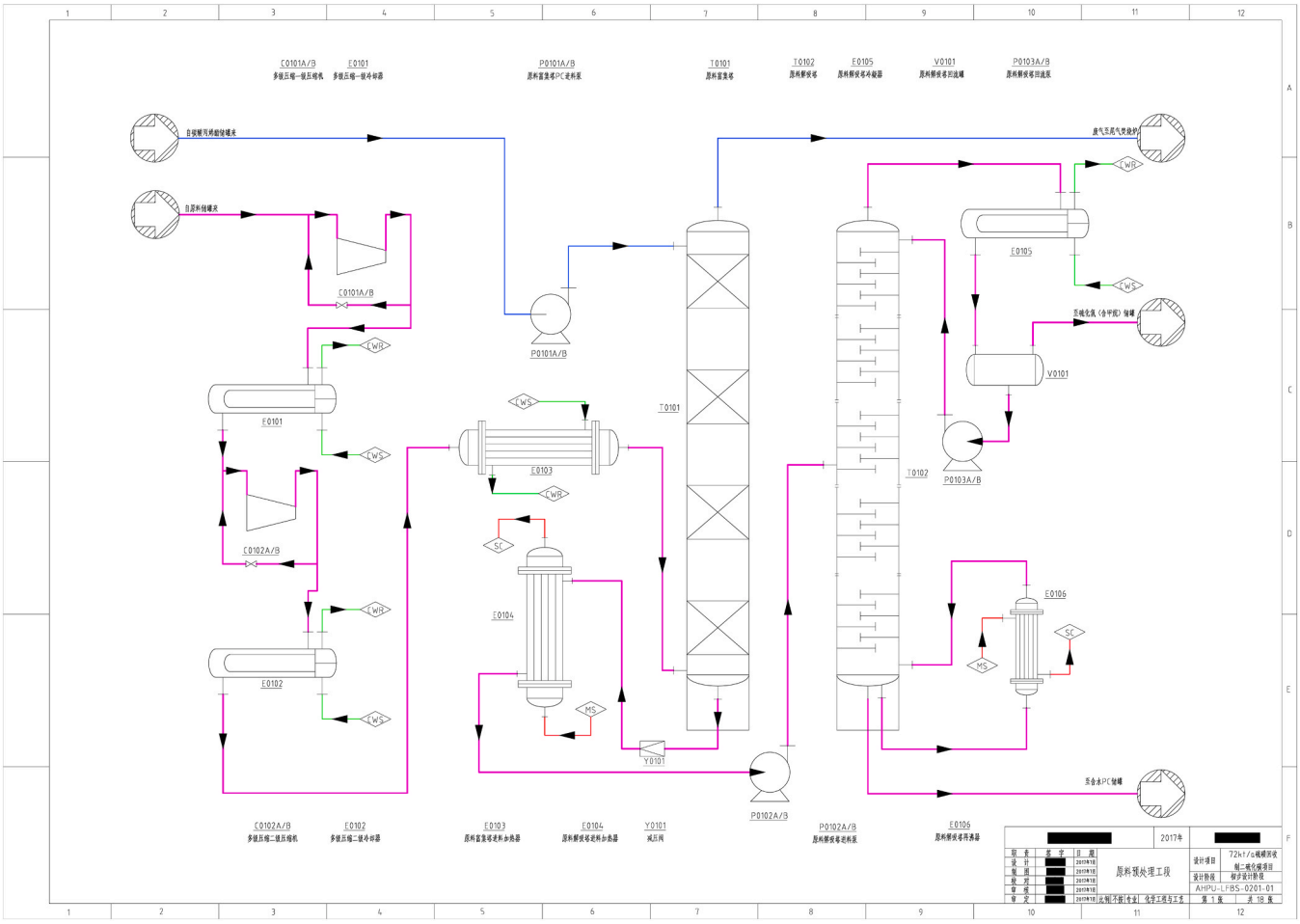


Fig. 8. An Example of the process flow diagram for sulfur recovery process.

Table 8
An example of the material balance sheet for an entire sulfur recovery process design.

Stream ID		S0101	S0245	S0246	S0309	S0418	S0425	S0429	S0430	S0322	S0403	S0410	S0434
Phase		Vapor	Vapor	Vapor	Vapor	Vapor	Liquid	Vapor	Vapor	Liquid	Liquid	Liquid	Vapor
Mass Flowrate/(kg/h)	**H_2S**	9610.96	0	0	0	0	0	0	0	0.000108	0.010966	2.97276	0
	SO_2	0	0	0	0	0	0	0	0	2.28×10^{-9}	0.031771	0	0.327856
	COS	0	0	0	0	0	0	0	0	7.41×10^{-12}	0.000323	0	0
	CS_2	0	0	0	0	0	0	0	0	0.000833	0.000266	0.003336	4.06×10^{-5}
	S_2	0	0	0	0	0	0	0	0	1.37×10^{-6}	1.69×10^{-5}	0	0
	S_6	0	0	0	0	0	0	0	0	0.238584	7.28×10^{-5}	0	0
	S_8	0	0	0	0	0	0	0	0	0.015546	1.66×10^{-5}	0	0
	CO	0	0	0	0	0	0	0	0	1.21×10^{-14}	1.09×10^{-5}	0.001954	1.640432
	H_2	0	0	0	0	0	0	0	0	1.21×10^{-15}	1.26×10^{-6}	0.000139	0.058215
	CO_2	15,082.68	0	0	105.1009	2061.417	0	7.041568	0	1.88×10^{-7}	0.006808	1.130046	18100.94
	N_2	0	15,278.58	0	45.15728	0	0	3.025456	10,972.73	1.63×10^{-9}	0.000127	0.022822	26299.47
	H_2O	492.7803	106.5295	0	0	0	0.002189	0	75.64221	0.000446	4076.064	5538.852	1784.919
	O_2	0	4060.824	3592.831	0	0	0	0	3331.29	0	1.51×10^{-12}	2.69×10^{-10}	772.694
	CH_4	63.49518	0	0	2329.616	0	0	156.08	0	1.25×10^{-8}	1.26×10^{-14}	2.19×10^{-12}	0.281179
	S	0	0	0	0	0	0	0	0	0	1.33×10^{-5}	0	0
	C_2H_6	0.512897	0	0	1.346438	0	0	0.090209	0	0	0	0	0.000439
	C_3H_8	0.028929	0	0	0	0	0	0	0	0	0	0	5.30×10^{-6}
	PC	0	0	0	0	0	30.99781	0	0	0	0	0	0.030739
Total Molar Flowrate/(kmol/h)		656.04	678.2197	112.2802	149.2578	46.83996	0.303754	10	500	0.00134	226.2569	307.5669	1473.444
Total Mass Flowrate/(kg/h)		25,250.45	19,445.93	3592.831	2481.22	2061.417	31	166.2372	14,379.66	0.255518	4076.115	5542.983	46960.36
Total Volume Flowrate/(m^3/h)		9943.25	16,303.77	2698.152	3827.988	1156.712	0.026036	256.4682	12,019.42	0.000241	4.161972	5.659689	51817.01
Temperature/℃		42.10	20	20	40	30	30	40	20	130	40	40	150
Pressure/kPag		171.33	0	0	0	0.2	2	0	0	298.675	−76.325	2398.675	−1.325

including hydrogen sulfide and sulfur dioxide. In some cases, this section may simply involve flaring or burning the tail gases to convert the remaining H_2S and SO_2 into SO_2 and SO_3, which are then safely released into the atmosphere.

However, sulfur recovery process also presents significant risks, particularly the potential leakage of toxic gases like H_2S, which not only increases explosion risks but also creates safety challenges due to the high temperatures involved. A notable incident occurred in 2001 in the UK, where uncontrolled mixing of waste materials during chemical waste treatment led to the release of H_2S gas, resulting in one fatality and multiple hospitalizations. This underscores the importance of conducting comprehensive hazard analysis. HAZOP analysis is particularly effective in identifying risks associated with waste mixing and gas release, helping to implement robust safety measures.

To evaluate the effectiveness of HAZOPCT, this study collects a comprehensive dataset consisting of 19 design works related to sulfur recovery processes. Each work includes P&IDs, PFDs, material balance sheets, equipment specification sheets, HAZOP analysis tables, and MSDSs for 18 commonly used chemicals. Table 6 presents an overview of the design works. What's more, a custom synonym table, which is shown in Table 7, is used to map variations in chemical names. To provide a representative view of the dataset, we present examples of the six types of data sources in Figs. 7 and 8, and Tables 8 through 11.

In the aforementioned 19 design works, there are significant similarities and differences in the main components of waste gases, products, absorbents, catalysts, scale, and various other aspects.

- **Waste gases:** Most of the 19 design works utilize sulfur-containing waste gases emitted from thermal power plants. These waste gases are primarily composed of nitrogen, oxygen, carbon dioxide, and SO_2, which are typical components of flue gases produced during combustion processes in power generation. However, design works 12 and 17 rely on waste gases from oil refineries that are rich in hydrocarbons and H_2S. Additionally, design works 1, 9, and 18 also utilize waste gases sourced from oil refineries, but these gases undergo tail gas incineration before processing. This incineration step results in H_2S and CO_2 as primary components, which are the key products following the oxidation of the sulfur compounds.
- **Products:** The types of products generated from various design works vary widely, including CS_2, H_2SO_4, sodium sulfate (Na_2SO_4), purified carbon dioxide (CO_2), and elemental sulfur (S), among others. This diversity reflects the differing approaches taken to treat waste gases.
- **Absorbents and catalysts:** Various design works utilize absorbents ranging from common sodium hydroxide (NaOH) and ammonia water ($NH_3{\cdot}H_2O$) to more specialized substances such as hollow fiber membranes, sodium phosphate (Na_3PO_4), and sodium bisulfite ($NaHSO_3$). Regarding catalysts, aluminum oxide (Al_2O_3), sodium molybdate (MoO_3), titanium dioxide (TiO_2) are the most commonly used chemicals in catalysts to facilitate the Claus process. However, some design works do not employ catalysts because they do not include the Claus process. Instead, they directly utilize the solution obtained from the acid gas sweetening process and obtain downstream products through sulfur post-treatment section.
- **Scale:** The annual sulfur recovery capacities of these processes vary significantly, ranging from 0.86 kilotons to 229.7 kilotons per year. Design works 9 (229.70 kt/y), 2 (166.59 kt/y), and 10 (103.26 kt/y) exhibit high sulfur recovery capacities, indicating that they are large facilities likely employing multiple stages of gas treatment. In contrast, design works 4 (0.86 kt/y), 14 (3.70 kt/y), and 7 (9.04 kt/y) represent less complex sulfur recovery processes.
- **Other aspects:** The types and configurations of stages in the reactors, along with operating temperature and pressure, energy consumption, and emission control measures, are also key factors to consider when conducting a HAZOP analysis. However, different

Table 9
An example of the equipment specification sheet for the heat exchangers used in sulfur recovery process.

Device ID	Category	Number	Connection Type	Specification	Installation	Material
E0103	Cooler	1	Single	BEM 900–4–271–4.5/19–1 I	Horizontally installed	AISI 304 stainless steel
E0104	Heater	1	Single	BEM 700–0.6–217.4–6/19–1 I	Vertically installed	AISI 304 stainless steel
E0105	Condenser	1	Single	BEM 700–0.6–120.9–4.5/25–2 I	Horizontally installed	AISI 304 stainless steel
E0106	Boiler	1	Single	BKM 1600–6.4–1343–9/25–4 I	Horizontally installed	AISI 304 stainless steel
E0201	Heater	2	Parallel connection	BEM 500–0.6–32.8–2/19–1 I	Horizontally installed	AISI 304 stainless steel
E0202	Cooler	2	Parallel connection	BEM 900–0.6–180.7–3/19–1 I	Horizontally installed	AISI 304 stainless steel
E0206	Heater	9	Parallel connection	BEM 600–0.6–38.5–2/25–1 I	Vertically installed	AISI 304 stainless steel
E0207	Cooler	1	Single	BEM 273–1.6–5–1.5/19–2 I	Horizontally installed	AISI 304 stainless steel
E0208	Heater	4	Parallel connection	BEM 400–0.6–11.5–1.5/25–1 I	Vertically installed	AISI 304 stainless steel
E2010	Cooler	3	Parallel connection	BEM 600–4–44.3–2/25–1 I	Horizontally installed	AISI 304 stainless steel
E0211	Cooler	1	Single	BEM 159–1.6–1.3–1.5/19–1 I	Horizontally installed	AISI 304 stainless steel
E0212	Cooler	1	Single	BEM 1000–0.6–176.5–3/25–1 I	Horizontally installed	AISI 304 stainless steel
E0213	Heater	1	Single	BEM 700–0.6–108.7–3/19–1 I	Vertically installed	AISI 304 stainless steel
E0214	Cooler	1	Single	BEM 500–1.0–32.8–2/19–1 I	Vertically installed	AISI 304 stainless steel
E0301	Cooler	1	Single	BEM 400–1.6–31.2–3/19–1 I	Horizontally installed	AISI 304 stainless steel
E0302	Heater	1	Single	BEM 900–1.6–361.4–6/19–1 I	Vertically installed	AISI 304 stainless steel
E0307	Heater	2	Parallel connection	BEM 500–0.6–32.8–2/19–1 I	Horizontally installed	AISI 304 stainless steel
E0308	Condenser	1	Single	BEM 700–0.6–194.1–6/19–4I	Horizontally installed	AISI 304 stainless steel
E0309	Boiler	1	Single	BEM 1200–0.6–657.9–6/19–1 I	Horizontally installed	AISI 304 stainless steel
E0401	Heater	3	Parallel connection	BEM 800–4–110–3/25–1 I	Horizontally installed	AISI 304 stainless steel
E0404	Cooler	1	Single	BEM 900–4–271–4.5/19–1 I	Horizontally installed	AISI 304 stainless steel
E0406	Condenser	2	Parallel connection	BES 1200–0.6–975.6–9/19–2 I	Horizontally installed	AISI 304 stainless steel
E0407	Boiler	1	Single	BEM1100–4–806.4–9/19–1 I	Vertically installed	AISI 304 stainless steel
E0408	Cooler	1	Single	BEM 1300–0.25–570.3–4.5/19–1 I	Horizontally installed	AISI 304 stainless steel
E0409	Cooler	2	Parallel connection	BEM 1800–2.5–2281.5–9/19–1 I	Horizontally installed	AISI 304 stainless steel
E0410	Condenser	1	Single	BEM 159–1.6–1.3–1.5/19–1 I	Horizontally installed	AISI 304 stainless steel
E0411	Boiler	1	Single	BEM 1400–6.4–729–6/25–1 I	Horizontally installed	AISI 304 stainless steel

design works exhibit significant variations in these aspects. For instance, design work 1 employs a multi-tubular reactor as the primary Claus reactor, whereas design work 2 utilizes a fixed-bed reactor.

3.1.2. Construction results of the HAZOP knowledge graph

Following the HAZOP knowledge graph construction framework, we integrated multi-source process safety information from 19 design works into a comprehensive HAZOP knowledge graph. The HAZOP knowledge graph encompasses 3743 entities, 70 types of relationships, and 9742 triplets. Specifically, the entities include 18 chemicals, 133 pieces of equipment, 165 streams, 223 process parameters, 388 process parameter deviations, 885 potential causes, 651 consequences, 945 safeguards, 176 controlling instruments, and 159 measuring instruments.

To evaluate the reliability of HAZOPCT, the HAZOP knowledge graph was partitioned into training and test sets using triples as the fundamental units. Since we are only interested in HAZOPCT's performance in predicting HAZOP analysis results, only triples that represent these conclusions are included in the test set. To put it specifically, we identified triples characterized by the relationships *"caused by"*, *"results in"*, or *"prevented by"* as indicative of HAZOP analysis results. Part of them were allocated to the test set, while the remaining triples, along with those that do not represent HAZOP analysis results, were assigned to the training set. Finally, the training set consists of 9282 triplets, with 2021 representing HAZOP analysis results, and the test set includes 460 triplets representing HAZOP analysis results. Consequently, the ratio of HAZOP analysis results in the training set to those in the test set is approximately 4.39:1.

3.2. Hyperparameter settings of HAZOPCT

In this study, the BERT pre-trained language model (Devlin et al., 2019) is adopted to generate semantic embeddings, comprising 12 layers, 768 hidden dimensions, 12 attention heads, and a total of 110 million parameters. And the KBGAT model (Nathani et al., 2019) utilizes two attention layers and two attention heads. The relevant hyperparameters of the KBGAT model (Nathani et al., 2019) and ConvKB model (Nguyen et al., 2018) are presented in Table 12.

3.3. Evaluation of HAZOPCT

Assuming the accuracy of all triplets in the test set, each triplet was compared to its corresponding prediction rankings produced by HAZOPCT, and the evaluation metrics include:

- MR (Mean Rank): Average rank of all triples in the prediction rankings.
- MRR (Mean Reciprocal Rank): Average reciprocal rank of all triples in the prediction rankings.
- Hits@K: Proportion of triples that are ranked among the top K predictions (K=1, 3, 10).

The model's performance in inferring causes, consequences, and safeguards for new accident scenarios is shown in Table 13. When evaluating only the top predicted result, HAZOPCT infers correct HAZOP analysis results in 70 % of cases. When considering the top ten predicted results, the accuracy rate increases to 86 %.

To further assess the effectiveness of the semantic embedding and graph attention mechanism, ablation studies were conducted. Comparisons were made with the CompGCN model (Vashishth et al., 2020), a composition-based graph convolutional network. Our findings indicate that HAZOPCT consistently outperforms CompGCN in HAZOP reasoning tasks across all evaluation metrics, underscoring the significance of incorporating self-attention mechanisms. It is noteworthy that different types of entities and relationships in HAZOP reasoning may possess varying levels of importance. For instance, equipment types may be more critical than their names in the context of HAZOP analysis. In contrast to CompGCN (Vashishth et al., 2020), which applies a uniform set of weight matrices to each relationship, HAZOPCT utilizes a graph attention network to dynamically compute attention weights. This allows for variability in the attention that each entity pays to its neighboring entities, thereby enabling the model to effectively capture the structural information inherent in the graph.

Furthermore, the performance of HAZOP reasoning, following the integration of semantic information, surpasses that of the inference conducted prior to this integration in terms of MRR, Hits@1, Hits@3, and Hits@10. It is important to note that MR's sensitivity to outliers and

Table 10
An example of the HAZOP analysis table for sulfur recovery process.

Name	Design Intention
72 kt/a Sulfur Recovery and Carbon Disulfide Production Project	Liquid sulfur from the storage tank is vaporized and mixed with methane-rich natural gas, reaching a feed temperature of approximately 500°C, and then fed into the carbon disulfide reactor

Parameter	Guide Word	Possible Causes	Consequences	Safeguards	Suggestions
Gas feed flow rate of reactor R0301	More	Upstream feed control valve FV−301 has malfunctioned or is opened excessively	Reaction rate accelerates, reactor temperature rises excessively, leading to potential explosion	- Safety valve SV−301 and rupture disk - High flow alarm FAH−301	Add a high flow interlock to shut down feed if flow exceeds safe limits
		Sudden increase in the flow rate from main plant	Reaction rate accelerates, reactor temperature rises excessively, leading to potential explosion	- Safety valve SV−301 and rupture disk - High flow alarm FAH−301	Add a high flow interlock to shut down feed if flow exceeds safe limits
	Less	Unintended malfunction of the control valve FV−301	Reaction slows, causing economic loss and inconsistent product quality	- Low flow alarm FAL−301	Add a low flow interlock to alert operators
		Feed pump P0301 has malfunctioned	Reaction slows, causing economic loss and inconsistent product quality	- Backup pump P0301B - Low flow alarm FAL−301	Automate switchover to backup pump P0301B upon detection of very low flow
		Unintended closure of the upstream pipeline	Reaction slows, causing economic loss and inconsistent product quality	- Low flow alarm FAL−301	Add a differential pressure sensor across pipelines to detect blockages
		Leakage from upstream equipment V0301/ V0302	Reaction slows, causing economic loss and inconsistent product quality	- Low-level alarms LAL−301 - Low flow alarm FAL−301	Add an interlock to stop feed if low level detected in V0301/ V0302
	None	Unintended closure of the control valve FV−301	Reaction stops, causing economic loss and inconsistent product quality	- Low flow alarm FAL−301	Add a low flow interlock to alert operators
		Feed pump P0301 has malfunctioned	Reaction stops, causing economic loss and inconsistent product quality	- Backup pump P0301B - Low flow alarm FAL−301	Automate switchover to backup pump P0301B upon detection of zero flow
		Unintended blockage of the upstream pipeline	Reaction stops, causing economic loss and inconsistent product quality	- Low flow alarm FAL−301	Add a differential pressure sensor across pipelines to detect blockages
		Leakage from the upstream equipment V0301/ V0302	Reaction stops, causing economic loss and inconsistent product quality	- Low-level alarms LAL−301 - Low flow alarm FAL−301	Add an interlock to stop feed if low level detected in V0301/ V0302
Temperature of reactor R0301	More	Upstream feed control valve FV−301 has malfunctioned and opened excessively	Excessive heat from exothermic reaction increases explosion risk; Side reactions cause coking	- Safety valve SV−301 and rupture disk - High temperature alarm TAH−301	Add a temperature interlock with cooling system
		Sudden increase in the flow rate from main plant	Excessive heat from exothermic reaction increases explosion risk; Side reactions cause coking	- Safety valve SV−301 and rupture disk - High temperature alarm TAH−301	Add a temperature interlock with cooling system
	Less	Control valve FV−301 has malfunctioned or is closed	Reaction slows, causing economic loss and inconsistent product quality	- Low temperature alarm TAL−301	Add low temperature interlock to alert operators or adjust heating
		Feed pump P0301 has malfunctioned	Reaction slows, causing economic loss and inconsistent product quality	- Backup pump P0301B - Low temperature alarm TAL−301	Automate switchover to P0301B upon detection of very low temperature
		Blockage or leakage of the upstream pipeline	Reaction slows, causing economic loss and inconsistent product quality	- Low temperature alarm TAL−301	Add flow monitoring to detect blockages early
		Leakage from the upstream equipment V0301/ V0302	Reaction slows, causing economic loss and inconsistent product quality	- Low-level alarms LAL−301 on V0301 and V0302 - Low temperature alarm TAL−301	Add interlock to stop feed if low level detected in V0301/ V0302
		Heat exchanger E0302 has malfunctioned	Reaction slows, causing economic loss and inconsistent product quality	- Temperature control system on E0302 (TIC−302) - Low temperature alarm TAL−301	Add interlock to adjust reactor conditions if E0302 malfunctions
Pressure of reactor R0301	More	Upstream feed control valve FV−301 has malfunctioned and is opened excessively	Safety valve SV−301 ruptures, halting reaction and shutting down system	- Temperature interlock with cooling system (TIC−301) - Safety valve SV−301 and rupture disk on reactor R0301	Add high pressure interlock to shut down feed or activate emergency cooling
		Sudden increase in the flow rate from main plant	Safety valve SV−301 ruptures, halting reaction and shutting down system	- Temperature interlock with cooling system (TIC−301) - Safety valve SV−301 and rupture disk on reactor R0301	Add high pressure interlock to shut down feed or activate emergency cooling

(*continued on next page*)

Table 10 (*continued*)

Name 72 kt/a Sulfur Recovery and Carbon Disulfide Production Project			Design Intention Liquid sulfur from the storage tank is vaporized and mixed with methane-rich natural gas, reaching a feed temperature of approximately 500°C, and then fed into the carbon disulfide reactor		
		Unintended blockage from the downstream pipeline	Safety valve SV−301 ruptures, halting reaction and shutting down system	- Temperature interlock with cooling system (TIC−301) - Safety valve SV−301 and rupture disk on reactor R0301	Add differential pressure monitoring across pipelines to detect blockages
	Less	Control valve FV−301 has malfunctioned or is closed	Reaction slows, causing economic loss and inconsistent product quality	- Low pressure alarm PAL−302	Add low pressure interlock
		Feed pump P0301 has malfunctioned	Reaction slows, causing economic loss and inconsistent product quality	- Backup pump P0301B - Low pressure alarm PAL−302	Automate switchover to P0301B upon detection of very low pressure
		Blockage or leakage of the upstream pipeline	Reaction slows, causing economic loss and inconsistent product quality	- Low pressure alarm PAL−302	Add flow monitoring to detect blockages early
		Leakage from the upstream equipment V0301/ V0302	Reaction slows, causing economic loss and inconsistent product quality	- Low-level alarms LAL−301 - Low pressure alarm PAL−302	Add interlock to stop feed if low level detected in V0301/V0302
		Leakage from the reactor R0301	Reaction slows, causing economic loss and inconsistent product quality	- Low pressure alarm PAL−302	Regularly inspect and maintain reactor R0301 to prevent leaks

its failure to emphasize top-ranked predictions make it a less reliable indicator of performance enhancements. The improvements observed in MRR and Hits@*K* metrics highlight that the incorporation of semantic information, which elucidates the meanings and contexts of nodes, facilitates a more profound comprehension by the inference algorithm. Consequently, this enhancement allows for effective inference capabilities when addressing sparse nodes, thereby improving the model's generalization abilities.

Overall, these findings highlight the potential application of HAZOPCT in HAZOP analysis table completion. For instance, Fig. 9 illustrates the consequence analysis results for the deviation scenario where the liquid level of T0103 is too high, as assessed by HAZOPCT. Notably, the correct label ranks first among all recommended historical conclusions. This indicates that for this specific deviation scenario, the output of HAZOPCT aligns with the actual HAZOP analysis results.

3.4. Enhancing HAZOP analysis completeness using HAZOPCT

In this section, we demonstrated the effectiveness of HAZOPCT in predicting previously unreported but plausible HAZOP analysis results. We utilized the entire existing HAZOP knowledge graph as input data for HAZOPCT, enabling it to predict missing triples that represent HAZOP analysis results.

During this process, HAZOPCT identified 14 overlooked yet reasonable conclusions, as shown in Table 14. For instance, HAZOPCT identified that an excessively high liquid level in distillation column T202, associated with the design work 9, would significantly reduce the heat exchange efficiency of the reboiler, thereby shortening its service life. Then the corresponding triplet can be added to the existing HAZOP knowledge graph. Therefore, it is obvious that HAZOPCT facilitates a more thorough and automated identification of potential hazards, ultimately contributing to safer industrial operations.

3.5. Example analysis of the local attention weight network

The attention mechanism employed in GATs reveals the importance of different features in the reasoning process, thereby substantially improving the interpretability of the model. We conducted an analysis of the self-attention weights in the final graph attention layer of the KBGAT model. As illustrated in Fig. 10, we utilized the subgraphs centered around pump 12-P0102 as an example, and presented the corresponding local attention weight network.

The results highlight that the model emphasizes key device attributes such as types, specifications, and measuring instruments, which are more relevant to HAZOP analysis. In contrast, less essential details, such as device names, receive lower attention. Notably, the shared attributes of similar devices enable HAZOPCT to align their embedding vectors effectively. This alignment facilitates knowledge transfer from prior analyses to new scenarios, resulting in consistent HAZOP analysis outcomes. Furthermore, the model emphasizes triplets that represent the results of HAZOP analysis with greater attention, aligning with the task of completing the HAZOP analysis table.

Once the attention weight network has been generated, both the inference results and the network can be presented to teams engaged in HAZOP analysis. This enables HAZOP analysis teams to gain a clearer understanding of the process safety elements emphasized by HAZOPCT during the inference of previously overlooked HAZOP analysis results. Therefore, the generated attention weight network can assist in verifying the reliability of the inference process.

4. Conclusions

In this paper, a HAZOP knowledge graph is constructed using a systematic methodology that includes the design of a HAZOP ontology, the extraction of relevant knowledge, and the integration of multi-source process safety information. Following the effective integration of all requisite information for HAZOP analysis into a cohesive knowledge graph, a completeness tool termed HAZOPCT is proposed to facilitate reliable completion of HAZOP analysis tables. The feasibility of this approach is substantiated by a case study centered on the sulfur recovery process. When evaluating the top ten predicted outcomes, HAZOPCT can infer valid HAZOP analysis results with an accuracy of 86 %. Additionally, an illustrative analysis of the local attention weight network elucidates the importance of each feature during the inference process of HAZOPCT, thereby enhancing the understanding of the decision-making basis for HAZOP reasoning within the model.

In future research endeavors, we intend to enhance HAZOPCT in several key areas. Firstly, we will concentrate on the development of intelligent methodologies for HAZOP analysis that utilize natural language generation techniques. Currently, the algorithm primarily serves as a reference tool, drawing on historical conclusions from similar process designs, which limits its ability to generate entirely new HAZOP analysis results for novel processes. Therefore, fully automating the analysis of project-specific issues remains a challenge. By incorporating natural language generation, we aim to overcome this limitation by allowing the system to generate context-sensitive recommendations that are more tailored to project-specific issues by training on project-specific data. This approach will enable HAZOPCT to produce more dynamic and

Table 11
An example of the material safety data sheets for sulfur recovery process.

Name	Hydrogen sulfide
Molecular Formula	H_2S
Molecular Weight	34.08
Health Hazards	• Hydrogen sulfide is a potent neurotoxin with strong irritant effects on mucous membranes. • Acute poisoning: Inhalation of high concentrations in a short period can cause symptoms like tearing, eye pain, foreign body sensation, photophobia, blurred vision, runny nose, throat burning, cough, chest tightness, headache, dizziness, fatigue, and confusion. • Some cases may exhibit myocardial damage, and severe exposure can lead to cerebral and pulmonary edema. • Exposure to extremely high concentrations (above 1000 mg/m^3) may result in sudden unconsciousness, respiratory arrest, cardiac arrest, and immediate death. • Eye exposure to high concentrations can cause conjunctival enema and corneal ulcers. • Chronic exposure to low concentrations may lead to neurasthenia and autonomic dysfunction.
Environmental Hazards	Hazardous to the environment, capable of polluting air and water
Fire and Explosion Hazards	• Highly flammable with strong irritant properties. • Forms explosive mixtures with air and can ignite when exposed to flame or high heat. • Reacts violently with strong oxidants, such as concentrated or fuming nitric acid, possibly causing explosions. • Being denser than air, hydrogen sulfide can accumulate in low areas and ignite upon contact with fire sources.
Hazardous Combustion Products	Sulfur oxides
First Aid Measures	• Eye Contact: Lift eyelids and thoroughly rinse with running water or saline for at least 15 minutes. Seek medical attention. • Inhalation: Move the victim to fresh air immediately. Maintain airway openness. Administer oxygen if breathing is difficult. If breathing stops, perform artificial respiration. Seek medical attention.
Firefighting Measures	• Firefighters should wear full fire-resistant and gas-tight suits and fight fires from an upwind position. • Cut off the gas source; if this is impossible, do not extinguish the flame at the leak. • Use water spray to cool containers and move them to open areas if possible. • Extinguishing agents: Water mist, alcohol-resistant foam, or dry powder.
Emergency Response	• Evacuate personnel from the affected area to an upwind location and isolate the area. For small leaks, isolate at least 150 m; for large leaks, 300 m. • Cut off ignition sources. Emergency responders should wear self-contained breathing apparatus and anti-static clothing. • Attempt to stop the leak from the upwind side, ventilate the area, and accelerate dispersion. • Dilute the gas with water mist and contain wastewater by building dikes or digging pits. • Vent leaked gas through a scrubber or fume hood, if possible, using ferric chloride solutions to neutralize it. • Properly handle leaking containers, repair, and inspect before reuse.
Handling Precautions	• Keep tightly closed, ensuring proper local and general ventilation. • Operators must receive specialized training and follow strict procedures. • Personal protective equipment: Use half-face respirators with gas filters, chemical safety goggles, anti-static clothing, and chemical-resistant gloves. • Avoid fire sources and smoking in the workplace. Use explosion-proof equipment and prevent gas leaks. • Prevent contact with oxidants and alkalis. • During transfer, ground and bond steel cylinders to prevent static electricity. • Handle cylinders carefully to prevent damage. • Keep fire extinguishing equipment and emergency leakage handling kits available.
Storage Precautions	• Store in a cool, ventilated warehouse away from heat sources and open flames. • Storage temperature should not exceed 30°C. Keep containers sealed and store separately from oxidants and alkalis. • Use explosion-proof lighting and ventilation systems. • Avoid equipment and tools that can generate sparks. • Prepare emergency response equipment for leaks.
Exposure Limits	• China MAC: 10 mg/m^3 • Former Soviet Union MAC: 10 mg/m^3 • OSHA TLV-TWA: 20 ppm, 28 mg/m^3 (ceiling) • ACGIH TLV-TWA: 10 ppm, 14 mg/m^3 • ACGIH TLV-STEL: 15 ppm, 21 mg/m^3
Monitoring Methods	Silver nitrate colorimetry
Engineering Controls	• Ensure strict containment and provide adequate local and general ventilation. • Install safety showers and eyewash stations.
Personal Protection	• Respiratory protection: Use gas filter masks if the concentration exceeds limits. In emergencies, use oxygen or air-supplied respirators. • Eye protection: Wear chemical safety goggles. • Body protection: Wear anti-static clothing. • Hand protection: Use chemical-resistant gloves. • Prohibit smoking, eating, and drinking in the workplace. Workers must shower and change clothes after shifts. Learn self-rescue and mutual aid techniques. • Supervision is required when entering confined or high-concentration areas.
Physical and Chemical Properties	• Main Components: Pure substance • Appearance: Colorless gas with a foul odor • Melting Point: −85.5°C • Boiling Point: −60.4°C • Vapor Density (Air = 1): 1.19 • Saturated Vapor Pressure: 2026.5 kPa (25.5°C) • Critical Temperature: 100.4°C • Critical Pressure: 9.01 MPa • Flash Point: −50°C • Autoignition Temperature: 260°C • Explosion Limits (V/V): 4.0 % - 46.0 % • Solubility: Soluble in water and ethanol • Primary Uses: Used in chemical analysis, such as for identifying metal ions.
Incompatibility	Strong oxidizers, alkalis
Acute Toxicity	LC50 (rat, inhalation): 618 mg/m^3
Environmental Impact	Harmful to the environment; avoid contamination of air and water.
Disposal Methods	Dispose of by incineration. Sulfur oxides released during incineration should be removed via scrubbing.
Transport Information	• Hazardous Goods No.: 21006 • UN Number: 1053 • Packaging Group: O52 • Packaging Methods: Steel cylinders, glass bottles in wooden crates. • Vehicles must carry fire extinguishing equipment and use flame arrestors on exhaust pipes. Avoid sparks during loading and unloading. • Follow railway hazardous goods loading regulations. Cylinders must have safety caps and wooden blocks to prevent rolling., laid flat, with consistent valve direction. • During stops, keep away from heat sources, designated routes and avoid residential areas. Avoid shunting during rail transport. • Transport during cooler periods to avoid sunlight exposure.

(*continued on next page*)

Table 11 (*continued*)

Name	Hydrogen sulfide
	• Prohibit mixed storage and transport with oxidants, alkalis, and food chemicals.

Table 12
Hyperparameters settings of HAZOPCT.

Hyperparameters of HAZOPCT	KBGAT	ConvKB
Batch size	9742 (All triplets)	128
Epoch	3600	200
Dropout ratio	0.3	0
Hidden unit	200	500
Learning rate	0.0005	0.0005
Weight decay	0.000005	0
Optimizer	Adam	Adam
Loss function	Margin Ranking Loss	Soft Margin Loss
Ratio of valid to invalid triples	2	40

relevant outcomes, helping the team address unique challenges in each project. Secondly, we aim to incorporate algorithms that can automatically extract process topologies and control structures from piping and instrumentation diagrams. In our previous research (Cui et al., 2010), we successfully achieved the automatic extraction of process information by integrating the intelligent HAZOP analysis tool with the chemical process design software SmartPlant P&ID. However, given that the adoption of SmartPlant P&ID software for process development has not been widespread across most process design projects, we will further explore the implementation of automatic extraction of process information from images or CAD drawings. This approach will facilitate broader applicability and ease of integration with existing design practices. Thirdly, we plan to conduct a comprehensive sensitivity analysis by directly perturbing input data (such as device types) and analyzing how these perturbations propagate to affect the HAZOP analysis results output by HAZOPCT, which will help us assess the sensitivity of the model and provide deeper insights into which features are most sensitive to changes in the input data. Lastly, we plan to integrate HAZOPCT into a next-generation HAZOP analysis system, thereby empowering HAZOP analysis teams to enhance the efficiency and quality of their analyses. By providing an integrated platform, we anticipate streamlining the HAZOP analysis process, reducing manual effort, and showing the analysis outcomes in a more user-friendly manner.

Statement

During the preparation of this work the authors used OpenAI tool only in order to improve readability of some sentences. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Zhao Jinsong: Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Li Zhiyuan:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Table 13
Performance comparison of reasoning methods on the HAZOP knowledge graph.

Method	Prediction	MR	MRR	Hits@1	Hits@3	Hits@10
CompGCN	Possible Cause	18.09	0.584	0.476	0.634	0.793
(Graph Convolutional Network)	Consequence	14.15	0.701	0.628	0.755	0.830
	Safeguard	13.91	0.650	0.561	0.688	0.805
	Total	15.44	0.637	0.544	0.683	0.806
HAZOPCT	Possible Cause	**7.18**	0.735	0.650	0.785	0.859
(Without Semantic Embedding)	Consequence	13.63	0.763	0.723	0.787	0.840
	Safeguard	**6.79**	0.739	0.675	0.768	**0.872**
	Total	**8.33**	0.742	0.676	0.778	0.860
HAZOPCT	Possible Cause	8.71	**0.743**	**0.669**	**0.798**	**0.871**
(With Semantic Embedding)	Consequence	**11.96**	**0.783**	**0.745**	**0.798**	**0.862**
	Safeguard	7.84	**0.756**	**0.704**	**0.778**	0.852
	Total	9.99	**0.757**	**0.7**	**0.789**	**0.861**

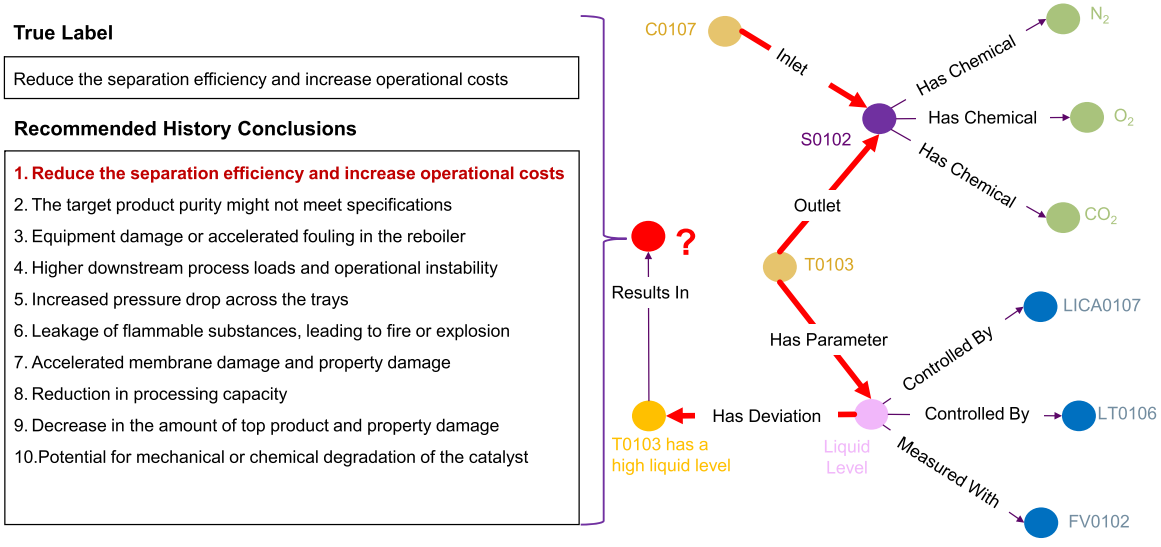


Fig. 9. An example of the results produced by HAZOPCT.

Table 14
HAZOP analysis results with supplementary insights from HAZOPCT.

Design work number	Parameter	Guide Word	Type of HAZOP analysis results	HAZOP analysis results (Bolded parts are supplemented by HAZOPCT)
4	Concentration of H_2S at the inlet of C201	More	Cause	Malfunction of the H_2S flow control valve, leading to excessive opening. Malfunction of the CO_2 flow control valve, leading to insufficient opening. Malfunction of the methanol flow control valve, leading to insufficient opening. **Instrumentation failure causing incorrect flow readings.**
	Feed flowrate of C201	Less	Cause	High resistance or blockage in the demister of the inlet separator. Blockage of the inlet filter. **Leakage due to rupture of the unit's inlet pipeline.**
8	Liquid level of T202	More	Cause	Malfunction of the outlet valve, leading to insufficient opening. Failure of the bottom pump resulting in shutdown. **Liquid level controller malfunction.**
			Consequence	Blockage of pipes and outlet valves, hindering material flow, affecting the normal operation. **The heat exchange efficiency of the reboiler will be severely reduced, impacting its lifespan.**
	Pressure of E208	More	Suggestion	Provide training on pressure management and emergency response. **Install a vent valve for non-condensable gases at the heat exchanger outlet.**
9	Pressure of R201	More	Cause	Malfunction of the pressure control valve on the inlet pipeline, leading to excessive opening. **Malfunction of the pressure control valve on the outlet pipeline, leading to insufficient opening.**
13	Outlet temperature of the heated stream of E501	More	Cause	Failure of the temperature control loop for the heating medium. Malfunction of the flow control valve for the heated stream, leading to insufficient opening. **Malfunction of the flow control valve for the heating medium, leading to excessive opening.**
		Less	Suggestion	Implement regular valve maintenance schedules. **Install a temperature interlock to regulate heating medium flow upon low temperature detection.**
15	Pressure of R201	More	Cause	Malfunction of the pressure control valve for the outlet pipeline, leading to insufficient opening. **Malfunction of the pressure control valve for the inlet pipeline, leading to excessive opening.**
			Suggestion	Perform routine equipment checks or replacements. **Add a pressure relief valve on the reactor top, with release directed to a flare system.**
17	Superheated steam flow of R201	More	Prevention	Install a temperature alarm interlock to regulate the superheated steam flow. **Set up an auxiliary emergency bypass on the superheated steam inlet pipeline.** **Install an explosion-proof safety valve on the reactor.**
	Temperature of R201	More	Consequence	Deactivation of the reactor catalyst, causing significant economic losses. **Excessive exothermic reactions could potentially lead to explosions.**
19	Temperature of R201	More	Consequence	Accelerated damage to the bipolar membrane, shortening replacement time and increasing costs. **Reduced electrodialysis capacity, leading to lower sulfur dioxide quality.**

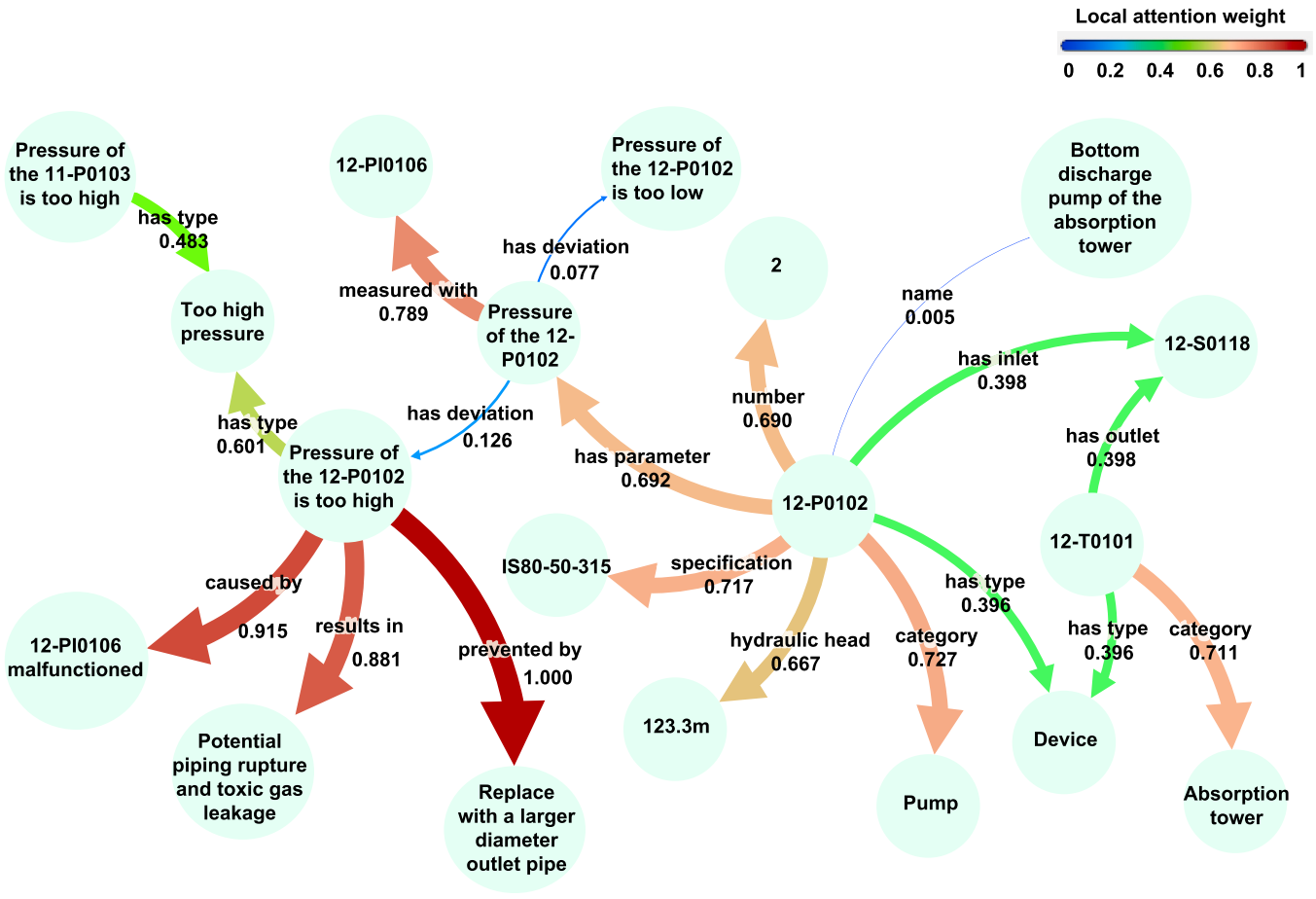


Fig. 10. An example of the local attention weight network.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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